



Prospect theory in the field: Revealed preferences from mutual fund flows[☆]

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ABSTRACT

Using mutual fund flows, we evaluate prospect theory with choice outcomes in the market. We provide strong support for prospect theory: under a standard set of parameters, funds whose past returns generate higher prospect theory value attract significantly larger future flows; we also find corroborative evidence using account-level data. Taking a revealed preference approach, we estimate the prospect theory parameters through a discrete choice model and find that our field-based estimates align well with previous experiment-based estimates. Moreover, we show that prospect theory offers a new framework for understanding flows, as it has explanatory power beyond existing drivers.

1. Introduction

Prospect theory has become a prominent alternative utility framework that describes investors' decision-making under uncertainty (Barberis and Thaler, 2003; Barberis, 2013). It was initially developed by Kahneman and Tversky (1979), Tversky and Kahneman (1992) to explain the choices observed in the laboratory, from which they derived a set of important parameters to govern the framework. While these parameters are widely adopted in the subsequent literature to explain various asset pricing phenomena, they are rarely confronted with choice outcomes outside of the laboratory, largely due to the scarcity of data.

In this paper, we examine the relevance of prospect theory for decision-making in the mutual funds market. First, we test whether prospect theory can explain the capital allocations by mutual fund investors. Specifically, we examine the link between mutual funds' prospect theory values under the standard parameters and fund flows.

Second, we conduct a revealed preference exercise to estimate the prospect theory preference parameters that best explain the fund flows. Lastly, we explore whether prospect theory provides a new framework for mutual fund flows, complementing the existing drivers identified in the literature.

Under prospect theory, investors mentally represent the gains and losses associated with each investment choice and evaluate them according to the value function in Tversky and Kahneman (1992) that assigns a prospect theory value to the given distribution of gains and losses. We adopt the empirical framework developed by Barberis et al. (2016) that relies on the distribution of past returns of mutual funds to represent gains and losses scenarios. This representation is natural and practical, since past returns are readily available in a wide range of investment choices. In addition, historical fund performance serves as the primary source of information that investors use to choose mutual funds.

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Prospect theory generates clear predictions for mutual fund flows. According to the theory, investors would prefer to allocate capital to funds with high prospect theory value. If prospect theory captures the behavior of a significant fraction of mutual fund investors, funds with high prospect theory values would attract larger investor flows, holding other factors constant. Consequently, our main hypothesis is a positive relationship between a fund's prospect theory value and its fund flows.

To test our hypothesis, we construct a measure, termed "TK" after Tversky and Kahneman, of prospect theory value for each mutual fund at a given time. Intuitively, TK captures the utility of holding the fund by a prospect theory investor. Empirically, we use a fund's preceding 60 months' realized returns to represent its return distribution.¹ We use the risk-free rate as the reference point to compute gains and losses.² We then feed the adjusted returns into the prospect theory value function to derive the TK value, using the standard prospect theory preference parameters in the literature.

Analyzing a large sample of actively managed equity mutual funds in the US, we find strong and robust evidence supporting the main hypothesis. Mutual funds with high prospect theory value experience significantly higher future fund flows. Our main results are based on panel regressions using future flows as the dependent variable and TK values as the independent variable. Our results remain robust after we control for various fund performance measures. Moreover, we employ several alternative econometric specifications to address potential biases arising from persistent regressors (Stambaugh, 1999). Our results remain consistent under these alternative specifications. We further examine the importance of specific components of prospect theory (loss aversion, convexity in the gain region versus concavity in the loss region, and probability weighting) for fund flows. We find that each of these components plays a significant role in explaining future fund flows.

Our analysis of fund-level flows captures the aggregate behavior of mutual fund investors. To provide a more granular analysis, we extend our investigation by utilizing data from a large retail brokerage that covers individual investors' positions and trading records. We test whether prospect theory explains individual investors' buying and selling decisions regarding mutual funds. Using account-level transaction data, we confirm that individual investors are more likely to hold and purchase mutual funds with high TK values. This indicates that individual investors, in line with prospect theory, exhibit a preference for funds that offer higher prospect theory values. Overall, our findings confirm that prospect theory plays a significant role in shaping investors' choices of mutual funds at both the aggregate level and individual level.

The analyses so far take the parameters governing the prospect theory preference at the standard values (based on experiments) in the literature. Another question that we seek to answer is what values of prospect theory parameters best fit the mutual fund choices of investors. To answer this question, we conduct a revealed preference analysis of fund choices by estimating a discrete choice model. We find strong evidence of loss aversion with a magnitude of 1.824, which falls between the traditional consensus of 2.25 for the loss aversion parameter (Tversky and Kahneman, 1992) and 1.31 proposed in recent lab-based studies (Walasek et al., 2018). Moreover, by leveraging the time variation in the loss aversion parameter from our quarter-by-quarter estimation, we document that prior investment performance negatively affects the degree of loss aversion, which supports the key assumption in Barberis et al. (2001).

¹ Common mutual fund data providers, e.g., Morningstar, display fund performance using graphs. The horizon can vary from one year to ten years, with five years being the most common.

² As shown in Section 4.1.4, our results are robust to alternative reference points such as zero, market returns, style benchmark returns, and expectation-based benchmark returns.

The estimated probability weighting parameters are 0.11 in the gain region and 0.23 in the loss region. These values are lower than those reported in lab-based studies, suggesting that mutual fund investors are even more subject to overweighting of small probabilities compared to participants in laboratory settings. The stronger effect of probability weighting observed in the field aligns with the preferences over lottery-like stocks (Kumar, 2009). Furthermore, we estimate the curvature of the S-shaped prospect theory value function to be 0.745, which is within the common range of findings from lab-based studies. Overall, our revealed preferences analysis provides strong support for the hypothesis that, when allocating capital among mutual funds, investors exhibit prospect theory preferences similar to those observed in laboratory settings.

To gain a deeper understanding of the underlying economic mechanisms, we examine how investors could intuitively assess TK when investing in mutual funds. In practice, investors might not explicitly calculate the TK of mutual funds. Instead, they can rely on more accessible signals that capture features of prospect theory, thus behaving as if they are chasing high-TK funds. One such signal is the loss-gain ratio of a fund, defined as the ratio of total losses (absolute value of the sum of negative monthly excess returns) to total gains (sum of positive monthly excess returns) over some historical window (e.g., the most recent 60 months). We show that the loss-gain ratio reliably predicts future fund flows, with patterns consistent with loss aversion.

We further investigate the factors driving the variations in TK from both the investor's perspective and the behavior of fund managers. From the investor's standpoint, the attractiveness of a fund is perceived through the lens of its past return distribution. Since TK is derived from the past return distribution of the fund, the statistical return moments naturally emerge as the key drivers. Our analysis shows that TK is positively affected by past cumulative returns, negatively associated with return volatility, and positively associated with return skewness. Furthermore, the loss-gain ratio is negatively related to TK. From the portfolio management side, we find that TK values across funds are tied to several holdings-based characteristics. Specifically, high-TK funds tend to hold high-TK stocks and exhibit risk-shifting behavior by the managers.

The extensive literature on mutual fund flows has identified many determinants of fund flows. Our analysis demonstrates that prospect theory provides significant incremental explanatory power compared to the existing rational and behavioral drivers. When compared with fund alphas, which reflect managers' skills, we find that TK exhibits distinct predictive power for fund flows. Another important driver of mutual fund flows is Morningstar ratings, with prior research consistently showing that investors tend to chase mutual funds based on these ratings (e.g., Reuter and Zitzewitz, 2021, Ben-David et al., 2022, and Del Guercio and Tkac, 2008). To ensure the robustness of our findings, we conduct an extensive set of robustness tests, incorporating Morningstar ratings as a control at both fund and share-class levels. Across all these tests, TK remains consistently and significantly positive. Moreover, we control for the Morningstar Risk Adjusted Return (MRAR), which reflects the expected utility value under power utility. The positive relation between TK and fund flow remains significant after including MRAR. It is worth noting that MRAR is also significantly and positively related to future fund flows. This finding suggests that mutual fund flows reflect the aggregate choices of investors consisting of both expected utility and prospect theory decision-makers.

We further distinguish prospect theory from several behavioral explanations of mutual fund flows. We first compare prospect theory to two prevailing behavioral patterns: return extrapolation and salience theory. We confirm that both have significant explanatory power for mutual fund flows, but the predictive power of prospect theory remains strong after controlling for these two behavioral drivers. This highlights the novel information that prospect theory delivers since all three variables are constructed using past mutual fund performance. The existing literature also relies on the maximum or skewness of fund

returns to capture mutual fund investors' lottery preferences. While lottery preference is consistent with prospect theory, our results demonstrate that TK offers additional predictive power for mutual fund flows beyond what is explained by lottery preferences.

Furthermore, we find that the significantly positive relation between TK and fund flows reflects non-fully rational behavior of mutual fund investors. First, the effect of TK on flows is stronger among retail investor-dominated funds and broker-sold funds, which are dominated by less sophisticated investors. This is consistent with the idea that retail and less sophisticated investors are more likely to display prospect theory preference. Second, the explanatory power of prospect theory for mutual fund flows strengthens during periods of high investor sentiment (as measured by the market sentiment index proposed by Baker and Wurgler, 2006) while it drops significantly during recessions. The significant interactions of investor sophistication and sentiment with the explanatory power of prospect theory for mutual fund flows indicate an element of bounded rationality underlying our main findings.

Finally, we find that although funds with high TK values tend to attract more investor capital, they do not subsequently outperform. In fact, we observe a negative relationship between TK value and future fund performance, further supporting the “dumb money” interpretation of capital flows to high prospect theory value funds. To delve deeper, we decompose fund flows into a TK-driven component (the part of fund flows that can be predicted by fund TK) and a non-TK-driven component (the part of fund flows unexplained by fund TK) by projecting fund flows on lagged TK. TK-driven flows significantly and negatively predict future fund performance, while non-TK-driven flows positively predict future fund performance. This suggests that mutual fund flows contain both rational and irrational components. Some investors can pick skilled funds, while other investors simply chase funds with high TK value.

Our paper contributes to the prospect theory literature that tests the theory with market data. Recent work along this line of research includes (Barberis et al., 2016), Barberis et al. (2021), Grinblatt and Han (2005) and Baele et al. (2019). A related literature evaluates prospect theory by linking investors' trading decisions to asset returns (An, 2016; An et al., 2020; An and Argyle, 2021). Diverging from previous research that primarily relies on market prices, our study emphasizes investors' demand and the choices they make when selecting mutual funds. This distinction is crucial because inferences drawn from choices and prices can differ significantly.³ Prices, as an equilibrium quantity, can be influenced by factors beyond demand. Consequently, relying solely on market prices may obscure inferences about the underlying preferences of investors. In contrast, choices directly reflect individuals' preferences. Thus, studying choices allows us to uncover and analyze investors' preferences more accurately. Our study provides strong support for prospect theory both at the aggregate level and at the individual level. Furthermore, this study obtains estimates of the prospect theory parameters using field data. The preference parameters from our revealed preference analysis confirm the important features of prospect theory preferences. Notably, the parameter estimates from our revealed preference analysis fall within a plausible range around the laboratory-based parameter values in the literature.

In addition, our study makes a valuable contribution to the existing literature on mutual fund flows. Since (Ippolito, 1992), it has been widely recognized that mutual fund investors tend to chase past performance. Recent studies in the flow literature, such as Barber et al. (2016), Berk and Van Binsbergen (2016), Song (2020), and Dannhauser and Pontiff (2024), investigate how different components of past performance affect fund flows. They show that both abnormal returns

(alpha) and factor-related returns predict future fund flows. However, this decomposition crucially depends on how investors assess risk, as highlighted in Jegadeesh and Mangipudi (2021). Our research takes a novel approach by examining mutual fund flows through the lens of prospect theory, which provides a psychologically realistic framework. We offer a new behavioral perspective to the flow literature and present empirical evidence supporting the significant incremental explanatory power of prospect theory for mutual fund flows.

Lastly, this study offers new evidence pertaining to the rationality of mutual fund investors. Guided by theory, we show that mutual fund investor behavior in aggregate is consistent with what prospect theory prescribes. Recent studies have also documented instances of non-fully rational behavior of mutual fund investors. For instance, they tend to chase growth stocks and experience poor subsequent performance (Franzoni and Schmalz, 2017); they persistently invest in high-fee mutual funds (Cooper et al., 2021). Additionally, mutual fund investors blindly follow *Wall Street Journal* rankings (Kaniel and Parham, 2017), Morningstar ratings (Del Guercio and Tkac, 2008; Ben-David et al., 2022), sustainability rankings (Hartzmark and Sussman, 2019), and are attracted by extreme past returns (Akbas and Genc, 2020). Our study adds to this body of research by highlighting the congruence between mutual fund investors' behavior and the predictions of prospect theory while controlling for other non-fully rational behavior within the mutual fund industry.

In two concurrent studies, Gu and Yoo (2021) and Guo and Schönleber (2022) also document that the fund's TK value is a positive and significant predictor of its future fund flow. Gu and Yoo (2021) contain only some basic findings and does not go in depth or explore the underlying mechanisms as our paper does. Guo and Schönleber (2022) differ from our paper both in focus and some details of the findings. Guo and Schönleber (2022) highlight the role of loss aversion, but they do not find empirical support for other features in prospect theory such as probability weighting or concavity/convexity,⁴ which is in strong contrast to both the original papers (Kahneman and Tversky, 1979; Tversky and Kahneman, 1992) and subsequent studies (Barberis, 2012; Barberis et al., 2016, 2021). We document that each feature of prospect theory independently and importantly contributes to explaining the choices of mutual fund investors. While (Guo and Schönleber, 2022) report that mutual funds prospect value does not reliably forecast future mutual fund returns, we find a significantly negative relation between prospect theory-driven fund flow and future fund performance. In terms of research focus, they examine the influence of prospect theory on both mutual fund investors and managers, while our study concentrates on mutual fund investors and presents more evidence on non-fully rational behavior among mutual fund investors. Moreover, unlike both papers, our study draws evidence from account-level data. We validate the role of TK value in driving investors' portfolio decisions on mutual funds, thereby providing further support for prospect theory at the micro-level. Furthermore, our paper provides a more extensive analysis by conducting a revealed preference analysis and estimating the prospect theory parameters using the mutual fund flow data.

The remainder of the paper is organized as follows. Section 2 provides the theoretical background and measure construction. Section 3 describes the data and sample. Section 4 presents results on the relationship between fund flows and prospect theory values of mutual funds as well as the calibrated parameters of prospect theory based on a revealed preference analysis. Section 5 demonstrates the relevance of prospect theory as a new framework for understanding mutual fund flows. Section 6 provides further evidence of the non-fully rational behavior of mutual fund investors. Section 7 concludes.

⁴ However, an earlier version of their working paper dated April 2021 stated (Page 3) “We analyze the individual building blocks of the prospect theory value (loss aversion, concavity and convexity, and probability weighting) and discover that the concavity and convexity feature, which means that the value function is concave over gains and convex over losses, plays an essential role in mutual fund flow prediction”.

³ For example, Bossaerts et al. (2022) show that, in a simple two-period dynamic model, prices in a myopic equilibrium are very close to those in a perfect foresight equilibrium, while the choices differ significantly.

2. Prospect theory valuation

Following Barberis et al. (2016), we break the decision-making process under prospect theory into two steps: “representation” and “valuation”. An investor first forms a mental representation of the distribution of gains and losses for a risky investment. The investor then evaluates this mental representation to assess the attractiveness of the investment.

2.1. Representation

Barberis et al. (2016) propose a mental representation of a stock's future prospect based on its past returns. In the mutual fund context, past returns are readily available to investors. For example, Morningstar, one of the major information sources for mutual funds, displays the performance of each mutual fund in a chart, commonly plotting recent fund returns at a monthly frequency going back five years. Thus, it is reasonable to use a fund's monthly returns over the preceding five years to represent possible future return scenarios.

For each mutual fund with at least 60 months of past returns at a given time, we sort the fund returns over the most recent 60 months in ascending order and obtain the following representation of the fund's gains and losses r (i.e., returns minus the reference point):

$$\left(r_{-m}, \frac{1}{60}; r_{-m+1}, \frac{1}{60}; \dots; r_{-1}, \frac{1}{60}; r_1, \frac{1}{60}; \dots; r_{n-1}, \frac{1}{60}; r_n, \frac{1}{60}\right), \quad (1)$$

where r_{-m} to r_{-1} are losses ordered from the most negative to the least negative, and r_1 to r_n are gains ordered from the smallest to the largest in magnitude, with each assigned an equal probability. For the main results reported in the paper, we use the risk-free rate as the reference point, following Barberis and Huang (2008). Our results are robust when we use other benchmarks such as the market or style returns when evaluating gains or losses.

2.2. Valuation

With the mode of representation in place, we describe the steps to calculate a fund's prospect theory value (termed “TK” henceforth) following Tversky and Kahneman (1992). For a general gamble

$$(x_{-m}, p_{-m}; \dots; r_{-1}, p_{-1}; r_0, p_0; r_1, p_1; \dots; r_n, p_n) \quad (2)$$

which delivers a payoff of x_i with a probability p_i , a prospect theory individual assigns the value

$$\sum_{i=-m}^n \pi_i v(x_i). \quad (3)$$

Here, $v(\cdot)$ is known as the value function, and π_i represents the decision weight for the payoff x_i . This decision weight is further determined by the probability weighting functions, $w^+(\cdot)$ and $w^-(\cdot)$:

$$\pi_i = \begin{cases} w^+(p_i + \dots + p_n) - w^+(p_{i+1} + \dots + p_n) & \text{for } 0 \leq i \leq n \\ w^-(p_{-m} + \dots + p_i) - w^-(p_{-m} + \dots + p_{i-1}) & \text{for } -m \leq i < 0. \end{cases} \quad (4)$$

To capture the key features of prospect theory, Tversky and Kahneman (1992) proposed the following functional forms:

$$v(x) = \begin{cases} x^\alpha & \text{for } x \geq 0 \\ -\lambda(-x)^\alpha & \text{for } x < 0 \end{cases} \quad (5)$$

and

$$w^+(P) = \frac{P^\gamma}{(P^\gamma + (1 - P)^\gamma)^{1/\gamma}}, \quad w^-(P) = \frac{P^\delta}{(P^\delta + (1 - P)^\delta)^{1/\delta}}, \quad (6)$$

where $\alpha, \gamma, \delta \in (0, 1)$ and $\lambda > 1$. Fig. 1 plots the value and probability weighting functions. Four features of prospect theory stand out. First, the carriers of value in prospect theory are gains and losses (x_i) – changes in wealth rather than final wealth levels – a concept known as *reference dependence*. Second, the value function $v(\cdot)$ is kinked at the

origin, indicating that the individual is more sensitive to losses than to gains of the same magnitude. This element of prospect theory is known as *loss aversion*, with the degree of the kink determined by the parameter λ . Third, $v(\cdot)$ is concave over gains but convex over losses. This feature, known as *concavity/convexity*, indicates that individuals have distinct attitudes towards risks in gain and loss regions. Finally, individuals use transformed probabilities derived from objective probabilities as the decision weights, as indicated by Eq. (6). The main consequence of the *probability weighting* feature is that individuals overweight the extreme and rare events.

Given the mental representation of a fund in Eq. (1), the TK value can then be computed following the prospect theory framework:

$$\begin{aligned} \text{TK} = & \sum_{i=-m}^{-1} v(r_i) \left[w^-\left(\frac{i+m+1}{60}\right) - w^-\left(\frac{i+m}{60}\right) \right] \\ & + \sum_{i=1}^n v(r_i) \left[w^+\left(\frac{n-i+1}{60}\right) - w^+\left(\frac{n-i}{60}\right) \right], \end{aligned} \quad (7)$$

where $v(\cdot)$, $w^-(\cdot)$, and $w^+(\cdot)$ are defined as above. Furthermore, we adopt the parameter estimates obtained by Tversky and Kahneman (1992) from experimental data:

$$\begin{aligned} \alpha &= 0.88, \lambda = 2.25, \\ \gamma &= 0.61, \delta = 0.69. \end{aligned} \quad (8)$$

We describe the detailed process to construct TK in Internet Appendix B.

3. Data and sample

3.1. Sample construction

Our main data source is CRSP's Survivor-Bias-Free US Mutual Fund Database. We focus on equity mutual funds, excluding ETFs/ETNs, variable annuities, and index funds.⁵ Our sample spans the period from January 1981 to June 2022. Fig. 2 plots the number of funds over the years. We start our sample period in 1981 to ensure more effective coverage of monthly Total Net Assets (TNA) and return data. Mutual fund data are recorded at the share-class level in CRSP. We aggregate share classes to the fund level following methods described in Berk and Van Binsbergen (2015). Most fund characteristics and performance metrics are value-weighted averages across share classes. Eventually, we select on average 2698 mutual funds per month for analysis. Fund TNA is the sum of share-class-level TNA, and age is based on the oldest share class. Fund flow, *Flow*, is computed following the standard method,

$$\text{Flow}_{i,t} = \frac{\text{TNA}_{i,t} - \text{TNA}_{i,t-1}(1 + r_{i,t})}{\text{TNA}_{i,t-1}}, \quad (9)$$

where $r_{i,t}$ denotes the return on fund i during month t and $\text{TNA}_{i,t}$ is the TNA value of fund i at the end of month t . Our findings are robust to an alternative measure of fund flows in Huang et al. (2011) computed as $\text{FLOW}_{i,t} = \frac{\text{TNA}_{i,t} - \text{TNA}_{i,t-1}(1 + r_{i,t})}{\text{TNA}_{i,t-1}(1 + r_{i,t})}$. We winsorize fund flows at the 5% and 95% levels.⁶ Factor loadings and alphas are computed using a rolling window of past-60-month fund returns.

⁵ Our main results are robust for both bond mutual funds and index mutual funds. See Table B5 in the Internet Appendix for details.

⁶ We apply this filter to mitigate the impact of measurement errors in fund flows (e.g., Elton et al. (2011) document that the dates on which fund mergers occur often differ from actual merger dates, which may generate extreme flow values). To ensure consistency, we also winsorize the other main variables at the 5th and 95th percentiles. Our main results are robust when we winsorize at the 1% and 99% levels (see Internet Appendix Table B4).

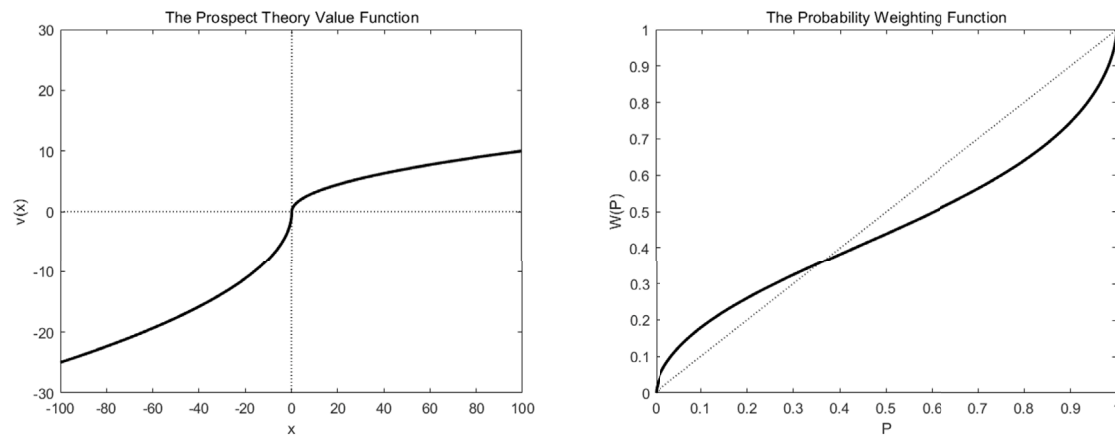


Fig. 1. Value Function and Probability Weighting Function in Prospect Theory

This figure depicts the value function and probability weighting function in prospect theory. The parameters for each function are based on the estimates reported in [Kahneman and Tversky \(1979\)](#).

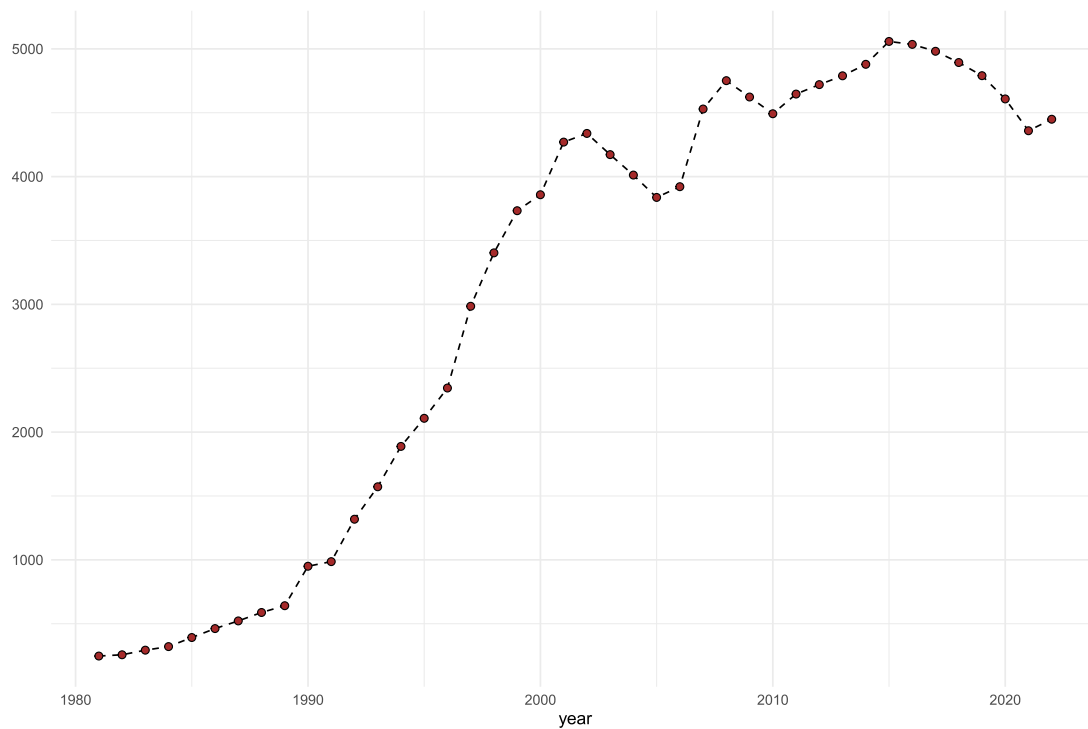


Fig. 2. Total Number of Funds Over Time

This figure graphs results indicating how the total number of funds varies over time. We use the CRSP Mutual Fund Database sample, which spans the period from January 1981 to June 2022. We focus on equity funds.

3.2. Summary statistics

[Table 1](#) Panel A reports the summary statistics of the main variables in our analysis. The summary statistics indicate that we have constructed a large and representative sample of mutual funds. The median fund size in our sample is \$258.5 million in total net assets (TNA). Active mutual funds on average experience small outflows and also have negative alphas in our sample period. In [Table 1](#) Panel B, we sort the funds in our sample into five groups each month by fund TK and report the average values of the variables labeled by the rows for each TK quintile. Measures of past performance, such as CAPM alpha, FF4 Alpha, and raw returns, all increase with TK value. High-TK funds also have higher TNA than low-TK funds, while they are similar in age. Most importantly, fund flow in the next period increases monotonically

with TK value. This hints at the presence of a positive relationship between TK value and fund flows.

4. Field evidence on prospect theory

We empirically examine the role of prospect theory in the market through three distinct approaches. First, we establish a connection between TK value and mutual fund flows which reflect the aggregate choice of mutual fund investors. Second, we delve deeper into the impact of prospect theory on the decision-making process of individual mutual fund investors using trading data at the account level. Moreover, to gain further insights, we employ a discrete choice model and conduct a revealed preference analysis. This enables us to directly infer mutual fund investors' preferences and estimate the prospect theory parameters that best fit the flow data.

Table 1
Summary statistics.

Panel A. Whole sample					
	mean	p50	sd	min	max
Flow	−0.002	−0.005	0.034	−0.364	0.932
TK	−0.033	−0.031	0.019	−0.093	0.004
Age	17.440	14.000	11.637	−17.000	99.000
TNA	1533.457	258.500	5799.611	0.097	199 057.300
Expense ratio (t−1)	0.013	0.012	0.013	−0.005	1.462
Turnover ratio (t−1)	0.822	0.540	2.015	0.000	150.910
CAPM Alpha	−0.086	−0.095	0.510	−6.928	4.759
FF4 Alpha	−0.123	−0.116	0.406	−7.382	4.514
Market loading	0.955	0.980	0.305	−2.881	3.801
SMB loading	0.148	0.056	0.344	−2.577	2.900
HML loading	−0.018	−0.020	0.348	−3.048	2.925
MOM loading	−0.015	−0.012	0.182	−3.347	2.439
FF4 R squared	0.812	0.883	0.190	−0.070	0.999
Return volatility	0.050	0.047	0.018	0.000	0.260
Cumulative returns (60 m)	0.547	0.470	0.610	−0.994	15.766

Panel B. Characteristics of High and Low TK Bins					
	Low TK	2	3	4	High TK
Flow	−0.007	−0.006	−0.004	−0.001	0.005
TK	−0.050	−0.038	−0.032	−0.027	−0.021
Age	16.570	17.186	17.559	17.955	17.933
TNA	718.650	1075.832	1310.737	1929.821	2634.391
Expense ratio (t−1)	0.016	0.013	0.012	0.011	0.011
Turnover ratio (t−1)	1.138	0.873	0.767	0.664	0.669
CAPM Alpha	−0.519	−0.207	−0.066	0.068	0.296
FF4 Alpha	−0.435	−0.203	−0.102	−0.019	0.146
Market loading	1.036	1.002	0.967	0.934	0.833
SMB loading	0.236	0.196	0.153	0.088	0.066
HML loading	−0.115	−0.061	0.001	0.034	0.051
MOM loading	−0.076	−0.009	0.000	0.001	0.012
FF4 R squared	0.725	0.813	0.851	0.864	0.808
Return volatility	0.066	0.052	0.047	0.044	0.040
Cumulative returns (60 m)	0.175	0.441	0.561	0.674	0.884

This table presents the summary statistics for the main variables in our analysis. We focus on mutual funds that exist for at least 60 months within our sample, and we aggregate data across share classes to get fund-level monthly observations. In Panel A, we report the summary statistics for the entire sample. In Panel B, we provide the average values of variables for each quintile, based on TK values formed on a monthly basis. CAPM Alpha and FF4 Alpha are in percentages. The sample spans from January 1981 to June 2022. Definitions of the variables are reported in Table A.18.

4.1. TK and fund flows

4.1.1. Portfolio sorts

Our main hypothesis is that prospect theory describes how investors make their choices among mutual funds: a mutual fund's TK value is positively related to its future flows. To test this relationship, we first utilize the standard portfolio sorting approach. One significant advantage of the portfolio sorting approach is its resilience against spurious regressions, as it depends solely on cross-sectional variations for estimation.

For each month from January 1986 through June 2022, we sort funds into deciles based on their TK values in the preceding month. We then examine the average fund flows for each decile in that month, both equal-weighted and value-weighted by the funds' TNA values. This yields a time series of average flows for each decile portfolio of funds sorted by TK. Lastly, we take the average flows over time and compare across deciles. The results are reported in Table 2. From the lowest TK decile to the highest TK decile, we observe a monotonic increase in the average fund flows, ranging from −0.4% to 0.9% when funds are equally weighted (Panel A) and from −0.6% to 0.8% when funds are TNA-weighted (Panel B). The negative (positive) inflows in the low (high) TK funds are both statistically significant. The spread in the monthly fund flow between the highest-TK funds and lowest-TK funds is around 1.4% and also statistically significant.

The results provide initial support for our main hypothesis that prospect theory affects mutual fund investors' decision-making when allocating capital across funds, as funds with high TK value tend to attract

significantly greater inflows. Although a univariate analysis provides an intuitive description of the relationship between TK and fund flows, it is not amenable to controlling for many other factors known to affect fund flows. In the following section, we employ regression analysis to control for various known determinants of fund flows.

4.1.2. Baseline regression

We now conduct regression analyses to test our main hypothesis. Specifically, we estimate the following regression model:

$$Flow_{i,t} = bTK_{i,t-1} + cX_{i,t-1} + \phi_i + \eta_t + \epsilon_{i,t}. \quad (10)$$

The dependent variable, $Flow_{i,t}$, represents the flow of mutual fund i in month t . Fund flows result from mutual fund investors' buying and selling decisions, i.e., the aggregate choices of mutual fund investors for each fund. Our variable of interest is $TK_{i,t-1}$, which measures a fund's prospect theory value based on the previous 60-month returns prior to t . $X_{i,t-1}$ is a vector of control variables that have been documented to predict fund flows.

Econometric Specification. Following the recent literature on mutual fund flows, we employ panel regressions with fixed-effects to estimate our baseline regression.⁷ We include both fund and time

⁷ For instance, Dannhauser and Pontiff (2024) use both fund and time fixed effects to study the flow–performance relationship for mutual funds. Similarly, Harvey and Liu (2019) incorporate both fund and time fixed effects to conduct panel regressions on the flow–performance relationship.

Table 2

Portfolio sorts: TK values and fund flows.

Panel A. Equally Weighted											
	TK Decile										
	Low	2	3	4	5	6	7	8	9	High	H – L
Flow	−0.004	−0.006	−0.004	−0.004	−0.002	−0.002	0.000	0.001	0.005	0.009	0.013
t-stat	(−2.50)	(−4.31)	(−3.69)	(−3.19)	(−2.35)	(−1.72)	(0.26)	(1.74)	(4.77)	(7.85)	(8.50)

Panel B. TNA-Weighted											
	TK Decile										
	Low	2	3	4	5	6	7	8	9	High	H – L
Flow	−0.006	−0.006	−0.004	−0.003	−0.002	−0.002	−0.001	0.001	0.004	0.008	0.014
t-stat	(−4.70)	(−4.28)	(−4.30)	(−2.92)	(−2.29)	(−1.63)	(−0.79)	(1.41)	(3.51)	(6.19)	(10.47)

This table presents the results from portfolio analysis on the relationship between funds' TK values and flows in the subsequent month. The sample spans from January 1981 to June 2022. Each month, we sort funds into deciles based on TK and calculate the average fund flows in the following month across all funds within each decile. We report the time-series mean of the average fund flows for each decile portfolio. For Panel A, we employ equal weights. For Panel B, we weight the funds by their total net asset value. The *t*-statistics are computed based on standard errors that are corrected by Newey–West with lags of 12 months.

fixed effects in Eq. (10). The fund fixed effects control for unobserved heterogeneity among mutual funds that is constant over time, such as time-invariant fund skills (Pástor et al., 2015) and diseconomy of scale (Zhu, 2018). Time-fixed effects help eliminate the effects of macroeconomic variables, market conditions, and regulatory changes.

Control Variables. The most important set of controls is fund-performance measures. Previous literature has documented extensive evidence that good past fund performance draws capital inflows from investors. We consider several types of performance measures. The first is cumulative fund returns in the preceding 60 months, which is a basic performance measure that investors can easily obtain. Since TK value is a nonlinear function of the preceding 60-month (adjusted) returns, we control for cumulative returns in the preceding 60 months to show that TK value contains additional information. The second performance measure is the fund alpha estimated using CAPM or Fama–French–Carhart four-factor model (Fama and French, 1993; Carhart, 1997). Alphas are known to predict future fund flows. Recently, several papers (e.g., Berk and Van Binsbergen, 2016 and Barber et al., 2016) have shown that CAPM alpha outperforms alphas under more complicated factor models in explaining fund flows. Lastly, we control for factor loadings on *mktrf*, *smb*, *hml*, and *umd* because investors respond to the portion of fund performance that is attributable to these factors (Barber et al., 2016).⁸

In addition to fund performance measures, we control for a battery of fund characteristics that investors might consider when selecting funds, including R-squared from the Four-Factor Model (as a proxy for fund manager skill following Amihud and Goyenko, 2013), fund age, size, expense ratio, and turnover ratio. We also control for fund-return volatility, which could diminish investors' ability to learn about fund-manager skills (Huang et al., 2022).

We estimate the regression model in Eq. (10) using fund flows as the dependent variable. By regressing flows on TK, we test whether prospect theory can explain investors' choices in aggregate. Table 3 presents the regression results. In all specifications, we include both fund and date fixed effects. All standard errors are two-way clustered at the fund and date levels. Column (1) reports the result of the univariate regression with TK as the only regressor, indicating a strong and positive relationship between TK and future fund flows, with a coefficient of 0.604 and a *t*-statistic of 26.89. The univariate regression result corroborates the sorting results reported in Table 2. When we add performance measures as control variables in Columns (2) and (3) of Table 3, TK remains positive and significant. Column (2) controls for the cumulative returns over the preceding 60 months and

CAPM alpha.⁹ The results are quantitatively similar if we use the alpha from the Fama–French–Carhart four-factor model (Carhart, 1997). In column (3), we additionally control for factor loadings under the Fama–French–Carhart four-factor model. Thus, prospect theory value provides distinct information about what drives fund flows beyond fund performance.

Table 3 column (4) includes additional fund characteristics as controls. The TK measure continues to be positive and significant, with a coefficient of 0.383 and a *t*-stat of 11.27. In economic terms, for an average fund in our sample with a TNA of \$1,533 million, a one-standard-deviation increase in TK maps into \$11.16 million in fund flows. We find that larger and older funds experience lower future flows, which is consistent in general with the theory of diseconomy of size. Other characteristics, such as expense ratios, turnover ratios, and fund activeness, are insignificant in the presence of TK value.

Overall, the findings reported in Table 3 strongly support our hypothesis. Prospect theory offers significant incremental explanatory power for fund flows beyond existing determinants including traditional performance measures and fund characteristics.

4.1.3. Robustness under alternative econometric specifications

The flow–TK predictive relationship arises from both cross-funds and within-fund comparisons. The univariate sorting result in Table 2 is based only on cross-sectional variations. The baseline regression in Table 3 is estimated from both the within- and cross-fund variations. Table 4 Panel A reports the results of three alternative specifications: panel regressions with time fixed effects only, fund fixed effects only and Fama–MacBeth regression. In column (1), we include only time-fixed effects. In this case the flow–TK relationship is purely estimated from cross-sectional variations. The TK coefficient is 0.30 with a *t*-stat of 9.30. This coefficient is a weighted average across periods of the slope estimates from the cross-sectional regressions in each period, with the weights being proportional to the number of observations and the sample variances in each cross-section. In column (2), only fund-fixed effects are included. The flow–TK relationship is estimated based on time-series variations (with a coefficient of 0.11 and a *t*-statistic of 3.64). This coefficient estimate essentially reflects a weighted average of fund-by-fund time-series regressions, with heavier weights assigned to funds exhibiting greater TK fluctuations over time.¹⁰ Column (3) reports the Fama–MacBeth regression, which is similar to column (1) when only the time fixed effects are included, except here the TK's coefficient is the equal-weighted average coefficients from period-by-period cross-sectional regressions.

⁸ To estimate the alphas and factor loadings, we use fund returns over the previous 60 months, which aligns with the estimation for TK. The results are similar if we use other estimation windows, such as 36 months.

⁹ The CAPM alpha's coefficient is positive and significant when we regress future flows on CAPM alpha alone, which is consistent with previous literature.

¹⁰ see proof details in Appendix B of Pástor et al. (2017)

Table 3
Prospect theory and fund flows.

	Dep. Var. Flow			
	(1)	(2)	(3)	(4)
TK	0.604*** (26.89)	0.343*** (12.88)	0.307*** (10.24)	0.383*** (11.27)
Cumulative returns (60 m)		0.006*** (6.16)	0.007*** (6.61)	0.008*** (7.05)
CAPM alpha		0.001 (1.06)	0.002** (2.28)	0.002* (1.91)
Market loading			−0.001 (−0.53)	−0.003 (−1.56)
SMB loading			0.003*** (2.99)	0.001 (0.88)
HML loading			0.000 (0.10)	0.000 (0.14)
MOM loading			−0.010*** (−8.43)	−0.008*** (−6.32)
FF4 R squared				−0.001 (−0.59)
Return volatility				0.114*** (3.91)
Ln(Age)				−0.013*** (−13.01)
Ln(TNA)				−0.003*** (−16.30)
Expense ratio (t−1)				0.001 (0.04)
Turnover ratio (t−1)				0.000 (1.35)
Adjusted R-Squared	0.124	0.128	0.130	0.138
N	871,549	861,927	861,927	859,562
Fund FE	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes
Controls	No	Yes	Yes	Yes

This table presents the predictive power of prospect theory in forecasting future fund flows. The values of prospect theory (TK) are estimated using fund returns from the preceding 60 months. The dependent variable is fund flow. In column (1), we run a univariate regression with TK as the only regressor. In column (2), we add the cumulative returns and CAPM alpha, both estimated using returns over the preceding 60 months. In column (3), we further control for loadings on factors in the Fama–French–Carhart four-factor model. In column (4), we include additional fund characteristics. In all specifications, we include both fund and date fixed effects. The definitions of variables are presented in Table A.18. The sample spans from January 1981 to June 2022. Standard errors are clustered at the fund and date level.

One potential concern when estimating the flow–TK predictive relationship is the persistence of the TK variable, which is constructed using a trailing window. We perform a panel unit-root test for TK using the Augmented Dickey–Fuller test (Maddala and Wu, 1999) and the Im–Pesaran–Shin test (Im et al., 2003). Both tests reject the null hypothesis that TK has a unit root in our panel setting.¹¹ To further alleviate this concern, we employ several different econometric specifications that are designed to address the persistent regressor problem. These results are reported in Panel B of Table 4. In column (1), we confirm the robustness of our baseline results using a first-difference estimator. As demonstrated in Hjälmarsson (2010), the bias caused by a persistent regressor in the time-series context is mitigated in the panel setting when the pooled estimator is used. However, when fixed effects are included, another bias emerges because the independent variable is demeaned using the full-sample time-series mean. This bias can be addressed using the recursive demeaning approach outlined in Hjälmarsson (2010) and Pástor et al. (2015). The key idea is to demean $flow_{i,t}$ and $TK_{i,t-1}$ by their mean values computed using fund i 's observation after time t (forward-demeaned), while using the $TK_{i,t-1}$ demeaned by the mean values computed using fund i 's observation

¹¹ The modified inverse χ^2 statistic in Augmented Dickey–Fuller test is 18.98 with a p -value close to zero. In Im–Pesaran–Shin test, the statistic Z_{ibar} is −10.17 with a p -value close to zero.

prior to t (backward-demeaned) as an instrumental variable for the forward-demeaned $TK_{i,t-1}$.¹² We follow this strategy and present our findings in column (2) of Panel B in Table 4. The instrumented forward-demeaned TK positively and significantly predicts fund flows. The third approach utilizes the bias correction method developed in Amihud and Hurvich (2004) to reduce the Stambaugh bias in multivariate time-series regressions. Specifically, we estimate the bias correction term as illustrated in equation (E3.2) in Amihud and Hurvich (2004) for each fund, and then estimate the fund-by-fund time series regressions with the bias correction term included. In column (3), we report the average value of the fund-by-fund TK coefficients and compute the t -statistic using the cross-sectional distribution of the TK coefficients. In all three approaches, the TK coefficients remain positive and significant.

4.1.4. Robustness with respect to variations in measurements of TK

Our results remain robust under alternative ways of calculating TK. When constructing a mental representation of a fund, we rely on monthly fund returns over the past five years. Internet Appendix Table B1 shows that, if we instead consider alternative windows of the previous one, two, three, four, or six years, we obtain similar results. Moreover, to address the concern that our results may be driven by the predictable flows when the last observation drops out from the previous reporting window, as documented by Phillips et al. (2016), we calculate TK using returns from non-standard windows of 55, 57, 59, 61, 63, and 65 months. As shown in Panel A of Internet Appendix Table B2, our findings remain robust. Additionally, we use TK calculated at the end of the previous year with rolling windows of one year or six years to predict the next year's annual flow. As reported in Panel B of Internet Appendix Table B2, the predictive power of TK remains significant.

Mutual fund investors may evaluate funds at a lower frequency than monthly. In Internet Appendix Table B3, we re-compute TK over a five-year window, but assume that investors evaluate gains and losses at either a quarterly or semi-annual frequency by compounding monthly returns into quarterly or semi-annual returns. Our results remain unaffected. Moreover, we consider alternative scenarios where investors evaluate portfolios of mutual funds instead of individual mutual funds. We document robust findings in Internet Appendix Table B10.

There is a lack of clear theoretical guidance regarding the specification of the reference point. In the main analysis, we use the risk-free rate as the reference point, a common choice in this literature. Our results remain robust to other alternative reference points. In Internet Appendix Table B6, we use market returns, style averages, and zero as the reference points, and our findings remain robust. In Internet Appendix Table B7, we adopt the idea of using expectations as reference points, following Kőszegi and Rabin (2006), Kőszegi and Rabin (2007, 2009). We construct four proxies for expectations: (a) average fund return over the past five years, (b) average style return over the past five years, (c) average expectation of existing investors across different cohorts, with the expectation in each cohort measured as the average return from the 60 months preceding their investment, and (d) extrapolation-based expectations following Barberis et al. (2015). Our findings remain robust under these alternative reference points.

We further consider alternative ways of representing a fund's return distribution using simulation methods and confirm that our findings are robust. Specifically, we conduct both non-parametric and GARCH-based simulations. For the non-parametric simulation, we simulate 1000 return observations based on a kernel density estimate derived

¹² Specifically, for any variable $v_{i,t}$ we define its forward-demeaned value as $\bar{v}_{i,t} = v_{i,t} - (\sum_{s=t+1}^T v_{i,s}) / (T - t)$, and the backward-demeaned value as $\underline{v}_{i,t} = v_{i,t} - (\sum_{s=1}^t v_{i,s}) / t$. With the demeaned values, we estimate a two-stage least squares regression. The first stage regression is $\overline{TK}_{i,t-1} = bTK_{i,t-1} + c\bar{X}_{i,t-1} + e_{i,t-1}$, where $X_{i,t-1}$ are other control variables. Then we use fitted values $\widehat{TK}_{i,t-1}$ from the first stage to run the second stage: $\overline{Flow}_{i,t} = \beta\widehat{TK}_{i,t-1} + \gamma\bar{X}_{i,t-1} + \varepsilon_{i,t}$. See Section 2.4 of Pástor et al. (2015) for more details.

Table 4
Prospect theory and fund flows: Alternative specifications.

Panel A. Fixed effects and Fama–MacBeth regressions			
Model specification:	Dep. Var. Flow		
	(1) Date FE	(2) Fund FE	(3) Fama–MacBeth
TK	0.301*** (9.30)	0.110*** (3.64)	0.210*** (5.61)
Adjusted R-Squared	0.068	0.110	0.112
N	859,592	859,562	859,592
Fund FE	No	Yes	
Date FE	Yes	No	
Controls	Yes	Yes	Yes

Panel B. Persistent regressor			
	Dep. Var. Flow		
	(1) First difference	(2) Recursive demean	(3) Amihud–Hurvich correction
TK (first difference)	0.643*** (3.79)		
TK (fitted value)		0.450*** (5.56)	
TK			0.632*** (5.97)
Controls	Yes	Yes	Yes
Adjusted R-Squared	0.122	0.033	0.29
N	852,324	848,518	837,790

This table presents the predictive power of prospect theory in forecasting future fund flows using different specifications. In Panel A, column (1) estimates the model with time-fixed effects but without fund-fixed effects. Column (2) estimates the model with fund-fixed effects but without time-fixed effects. Column (3) estimates the model using Fama–MacBeth regression. In Panel B, we adopt several alternative econometric specifications designed to address the persistent regressor problem. In column (1), we use the first-difference estimator. In column (2), we follow the recursive demeaning approach outlined in [Hjalmarsson \(2010\)](#). In column (3), we address the Stambaugh bias by following the approach in [Amihud and Hurvich \(2004\)](#). In all columns of both panels, the control variables are identical to those included in column (4) of [Table 3](#). The sample spans from January 1981 to June 2022. Standard errors are clustered at the fund and date level.

from the trailing 60-month returns. We then compute TK using the simulated returns. For the GARCH-based simulation, we fit a GARCH model with an expanding window for each fund and then simulate 1000 sample paths for returns from the past five years. We compute a TK value and then average it across 1000 simulations. Internet Appendix Table B8 shows that TKs from both simulation methods significantly predict fund flows.

4.1.5. Dissecting TK: The importance of individual features of prospect theory

Prospect theory synthesizes four behavioral traits associated with individual decision-making — reference dependence, loss aversion, concavity/convexity, and probability weighting. *Reference dependence* involves selecting the appropriate benchmark (i.e., reference point) for calculating gains and losses. *Loss aversion* suggests that an individual is more sensitive to losses than to gains of the same magnitude. *Concavity/Convexity* describes the feature that the prospect theory value function, $v(\cdot)$, is concave over gains but convex over losses. *Probability weighting* involves a situation in which an individual uses transformed probabilities when evaluating a gamble. The main consequence of probability weighting under cumulative prospect theory is inflated probability of tail events.

To gain insights on how these individual features affect mutual fund investor choices, we construct alternative prospect theory values that focus on only one feature by turning off others. We achieve this by applying alternative parameters to Eq. (7). To construct the prospect theory value with the loss-aversion component only, we use the parameter set $(\alpha, \gamma, \delta, \lambda) = (1, 1, 1, 2.25)$, instead of $(0.88, 0.61, 0.69, 2.25)$ in the original TK. Similarly, the prospect theory value featuring only concavity/convexity is computed using parameters $(\alpha, \gamma, \delta, \lambda) = (0.88, 1, 1, 1)$.

Table 5
Prospect theory features and fund flows.

	Dep. Var. Flow				
	(1)	(2)	(3)	(4)	(5)
LA	0.839*** (14.17)			0.415*** (5.77)	
CC		0.974*** (15.12)		0.598*** (7.78)	
PW			0.623*** (9.59)	0.233*** (3.55)	
TK					0.383*** (11.27)
Adjusted R-Squared	0.139	0.140	0.137	0.140	0.138
N	859,562	859,562	859,562	859,562	859,562
Fund FE	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes

This table presents the predictive power of each individual feature of prospect theory in forecasting future fund flows. The dependent variable is fund flow, and the specifications vary depending on which components of prospect theory are incorporated into prospect-theory values. For explanatory variables in columns (1) through (3), we calculate prospect-theory values using only loss aversion, concavity/convexity, and probability weighting, respectively. In column (4), we include all three independent features in one regression. In column (5), we replicate our main specification and use TK as the explanatory variable. All values are estimated using fund returns from the preceding 60 months. In all specifications, we include both fund and date fixed effects. The control variables are identical to those included in column (4) of [Table 3](#). The sample spans from January 1981 to June 2022. Standard errors are clustered at the fund and date level.

The prospect theory value with probability weighting only is calculated using parameters $(\alpha, \gamma, \delta, \lambda) = (1, 0.61, 0.69, 1)$.

We re-estimate the baseline regression Eq. (10) using loss aversion, concavity/convexity, and probability weighting individually in place of

the original TK that incorporates all the features. The results are reported in the first four columns of Table 5. The fifth column repeats our baseline result in column (4) of Table 3. Columns (1) through (3) report the regression results for each feature individually. All coefficients are positive and statistically significant. In column (4), we include all three individual features in one regression and find that each feature still has strong predictive power for fund flows. We conclude that each element of prospect theory has a significant and independent contribution to the explanatory power of prospect theory for fund flows.

Barberis et al. (2016) find that probability weighting is the primary driver for explaining the cross-section of expected stock returns. In their setting, if the probability-weighting feature is turned off, the standalone predictive power of loss aversion and concavity/convexity drops markedly. The discrepancy between the two sets of results is attributable to the different outcome variables used in the two papers. Expected stock returns, the outcome variable in Barberis et al. (2016), are jointly determined by investor preferences and other variables. By using a direct measure of investors' choices, a simpler outcome variable, this paper can potentially provide a sharper test of investor preference.

In a subsample analysis, we find that the relative importance of TK components systematically varies across fund styles. Specifically, loss aversion significantly predicts flows for value and large-cap funds but not for growth funds, while probability weighting significantly drives flows for growth and small-cap funds but not for value or large-cap funds. These findings, reported in Internet Appendix Table B11, indicate the alignment between various features of prospect theory and different styles of mutual funds. These results are consistent with the findings of Polkovnichenko et al. (2019), who demonstrate that value fund investors are more concerned with downside risk, while growth fund investors are more oriented towards upside potential (captured by probability weighting function). Polkovnichenko et al. (2019) estimate probability weighting function parameters using prices of S&P 500 index options and utilize the risk-neutral distributions of returns to construct empirical measures of investor preference for upside potential and downside protection. Using different methodologies, both papers show that the features of investor preferences that drive fund flows are consistent with non expected utility models emphasizing higher order moments of fund performance, thus making the case for an important role played by behavioral preferences in demand for actively managed mutual funds.

4.1.6. New subscription versus redemption

While the fund flow variable captures aggregate mutual fund investor choices, it masks potential differences between inflows and outflows. Chevalier and Ellison (1997) show that inflows are more responsive to good performance than outflows are to poor performance. This hints that an important difference may exist when mutual fund investors make purchase and sale decisions.

Using more detailed flow data from the CRSP Mutual Fund Database, we construct monthly inflow and outflow measures using the “new subscription” and “redemption” variables from the database, respectively.¹³ Several months of the sample contain significantly fewer records or are entirely missing data for new subscriptions and redemptions. Therefore, we have implemented a restriction requiring a minimum of 1000 funds with non-missing data for each month. We set new subscription and redemption values to be missing if they exceed 100% of the fund's TNA and winsorized these values at the 1% level. We scale the dollar amount of new subscription and redemption by the fund size at the previous month's end. We then estimate the regression specified in Eq. (10), using new subscription or redemption separately as the dependent variable.

¹³ These data are collected from SEC mutual fund filings and are therefore available only after 1993. The link table between flow data and fund return data begins only in July 2003.

Table 6

Prospect theory and alternative measures of fund flows: New subscriptions and redemptions.

	New Subscriptions	Redemptions
	(1)	(2)
TK	0.214*** (2.31)	-0.047 (-0.43)
Adjusted R-Squared	0.485	0.526
N	312,567	312,567
Fund FE	Yes	Yes
Year FE	Yes	Yes
Controls	Yes	Yes

This table compares the predictive power of prospect theory in forecasting future funds' new subscriptions and redemptions. The values of prospect theory (TK) are estimated using fund returns from the preceding 60 months. The dependent variables are new subscriptions and redemptions, which are scaled by fund size to reflect fund inflows and outflows, respectively. In column (1), we present the predictive power of prospect theory for future new subscriptions. In column (2), we present the predictive power of prospect theory for future redemptions. We require a minimum of 1000 funds with non-missing data for each month, and set new subscription and redemption values to be missing if they exceed 100% of the fund's TNA and winsorized these values at the 1% level. In all specifications, we include both fund and date fixed effects. The control variables are identical to those included in column (4) of Table 3. The sample spans from July 2003 to June 2022. Standard errors are clustered at the fund and date level.

Table 6 presents the results. For both new subscription (column (1)) and redemption (column (2)), we estimate the regression with all the control variables included (but for the sake of brevity, we do not report their coefficients). Column (1) shows that TK significantly and positively predicts new subscriptions. However, the TK coefficient becomes insignificant when redemption is the dependent variable. It suggests that mutual fund investors are more responsive to funds' prospect theory value in their purchase decisions than in their selling decisions.

4.2. Account-level evidence

Our main analyses infer investor preference for mutual funds based on the aggregate demand across investors captured by mutual fund flows at the fund level. In this section, we use individual account-level positions and transaction data from a large retail brokerage to test whether investors display prospect theory preferences.¹⁴ The account-level tests provide us with more granular examinations of our hypothesis but the sample period is much shorter due to data limitation. We first estimate the following regression,

$$AmtHeld_{i,j,t} = \beta TK_{j,t-1} + \gamma X_{j,t} + \lambda_i + \eta_t + \varepsilon_{i,j,t}, \quad (11)$$

where $AmtHeld_{i,j,t}$ is investor i 's amount of holdings in fund j on date t . We compute two measures of holdings: dollar amount held as a proportion of the overall account value, or as a proportion of a fund's TNA. $X_{j,t}$ is a set of control variables same as our baseline regression in Table 3, except all the variables are constructed at share-class level. λ_i and η_t are account and date (year-month) fixed effects. By including account and date fixed effects, we explore variations within one account while controlling for time-specific shocks. This enables us to determine how TK values of funds affect investors' allocation decisions.

In addition to holdings, we also examine whether investors' trading patterns fit the description of prospect theory. We estimate a model that is similar to (11), but with an alternative dependent variable, “NetBuy”,

$$NetBuy_{i,j,t} = \beta TK_{j,t-1} + \gamma X_{j,t} + \alpha_i + \eta_t + \varepsilon_{i,j,t}, \quad (12)$$

where $NetBuy_{i,j,t}$ is the net amount that investor i trades in fund j in month t . $TK_{j,t}$ is the TK value for fund j in month $t - 1$. Similar to holdings, we consider two transaction measures: the ratio of dollar amount of trading to the account value or to the fund size.

Table 7
Prospect theory and mutual fund holdings and transactions: Account-level evidence.

Panel A. TK and holdings		
	(1) Amt Held/Balance (%)	(2) Amt Held/Fund size (bps)
TK	45.357*** (4.73)	1.037*** (6.85)
Adj. Rsq.	0.839	0.755
N	1,316,974	1,519,974
Acct FE	Yes	Yes
Date FE	Yes	Yes
Controls	Yes	Yes

Panel B. TK and Transactions		
	(1) NetBuy/Balance (%)	(2) NetBuy/Fund size (bps)
TK	6.260*** (3.95)	0.089*** (4.67)
Adj. Rsq.	0.093	0.098
N	1,368,438	1,513,620
Acct FE	Yes	Yes
Date FE	Yes	Yes
Controls	Yes	Yes

This table presents the predictive power of prospect theory in explaining mutual fund investors' portfolio decisions. Specifically, we focus on account-level positions and transactions from a large brokerage firm spanning from 1991 to 1996 (see Barber and Odean (2000) for details). The values of prospect theory (TK) are estimated using fund returns (share-class level) from the preceding 60 months. The dependent variable in Panel A is the dollar amount of mutual fund holdings in an account scaled by account balances (*Amt Held/Balance*, in percent) or by fund size (*Amt Held/Fund Size*, in basis point). The dependent variable in Panel B is the net dollar amount transacted for a given fund in an account from the preceding month to the current month scaled by the account balance (*NetBuy/Balance*, in percent) or by fund size (*NetBuy/Fund Size*, in basis point). We include the same set of control variables as in column (4) of Table 3, except all the variables are constructed at the share-class level. In all specifications, we include both fund and date fixed effects. Standard errors are clustered at the account and date level.

Panel A of Table 7 reports the results of regression (11). For both holding measures, we find a positive and significant coefficient of TK. Thus, individual investors' portfolio choices are consistent with prospect theory. In terms of economic magnitude, the coefficient of column (1) indicates that one standard deviation increase in TK value would boost the investor's holding of that fund in their portfolio by 6.8%. Panel B reports the results of regression (12). The coefficient of TK is consistently positive and significant, suggesting that prospect theory value has a robust and significant explanatory power for individual investors' mutual fund choices. Using the coefficient in column (1) as an example, an investor would invest 0.94% more of their account value in a fund when the fund's TK increases by one standard deviation. Overall, the findings based on individual investors' portfolio decisions regarding mutual funds are consistent with the patterns we document using fund-level flow data.

4.3. Revealed-preference analysis

Thus far, our computation of TK value has been based on the following parameters:

$$\alpha = 0.88, \lambda = 2.25$$

$$\gamma = 0.61, \delta = 0.69$$

While these parameters are widely utilized in the finance literature (Barberis et al., 2016), they are based on laboratory settings and may not fully capture the preferences of real-world investors. In this section, we conduct a *revealed preference* analysis based on the field data from mutual funds investment. To this end, we employ a discrete choice model to determine which set of parameter values from prospect theory best explains the investment choices made by mutual fund investors.

4.3.1. A discrete choice model

Specifically, investors in our discrete choice model are presented with a "product space" consisting of $J + 1$ funds. Fund 0 is the baseline category, proxied by the Vanguard index fund (Roussanov et al., 2021). The remaining J funds are a collection of actively managed domestic equity funds, each indexed from $i = 1$ to J . When making investment decisions, investors select a fund i from both the index fund 0 and the remaining J active funds. Their choices are guided by the following utility function:

$$\delta_i = bTK_i(\theta, R_i) + c_k \sum_k x_i^k + e_i \quad (13)$$

Here, the TK value of fund i , denoted as TK_i , is jointly determined by prospect theory parameters $\theta = [\alpha, \lambda, \gamma, \delta]$ and the historical returns R_i of fund i . The characteristics of fund i , such as its size, turnover ratio, and expense ratio, are denoted by x_i^k . The term e_i represents the investor's idiosyncratic utility associated with fund i , which remains unobservable to researchers. We follow the standard industrial organization (IO) literature (Berry, 1994) and assume e_i follows a type I extreme distribution. Consequently, the probability of an investor selecting fund i is determined as $Prob_i = e^{\delta_i} / \sum_{j=0}^J e^{\delta_j}$, which is essentially the multinomial logit model.¹⁵

4.3.2. Estimation

We estimate the model in (13) using investors' choices among the retail equity funds.¹⁶ Specifically, we use the new subscriptions to

¹⁵ We set the utility of the baseline category as zero, i.e., $\delta_0 = 0$. The probability of choosing the index fund and all other funds can be rewritten as

$$Prob_0 = \frac{1}{1 + \sum_{j=1}^J e^{\delta_j}}, \quad Prob_i = \frac{e^{\delta_i}}{1 + \sum_{j=1}^J e^{\delta_j}}$$

¹⁶ See Barber and Odean (2000) for more details about the data.

capture the investor's purchase decision making process. This data is from the CRSP Mutual Fund Database. However, the data quality in the initial years is inadequate, characterized by a substantial number of missing values. As a result, our estimation sample is limited to the quarters from 2013 to 2022 that have more than 1000 non-missing observations.

In each quarter, we observe the inflows (new subscriptions) to each fund, denoted as f_i . This enables us to calculate the quarterly "market share" as $s_i = f_i / \sum_{j=0}^J f_j$. Utilizing the market share, we can estimate the parameters θ by maximizing the following likelihood function:

$$\ln L = \ln \prod_{j=0}^J \text{Prob}^{s_j} \quad (14)$$

Specifically, we run a MLE in each quarter based on the above equation. Then we obtain the time series of estimated parameters, $\hat{\theta}_t = [\alpha_t, \lambda_t, \gamma_t, \delta_t]$.

4.3.3. Implied prospect theory parameters from fund flows

In Panel A of Table 8, we report the averages, the standard errors, and the confidence intervals of the estimated $\hat{\theta}_t$ over the quarters. Overall, the revealed parameters in the field from our structural model provide strong support for the features of prospect theory. We begin by examining the degree of loss aversion, denoted as λ , which is a prominent feature of prospect theory. The original work by Tversky and Kahneman (1992) has widely influenced the belief that the degree of loss aversion is approximately 2. However, recent studies based on more representative samples of participants have shown that the true level of loss aversion may be lower than 2. In our findings, the revealed degree of loss aversion from mutual fund flows averages out to be 1.824, falling between the magnitude of 2.25 suggested in Tversky and Kahneman (1992) and the value of 1.31 reported in recent studies, such as Walasek et al. (2018). Our approach yields an average estimate of 0.745 for the parameter α that determines the S-shape of the prospect theory value function. This is in line with the experimental studies that typically produce α estimates ranging from 0.5 to 0.95.

Another important feature in prospect theory is probability weighting. The parameters γ and δ represent the weighting schemes in the gain and loss regions, respectively. According to prospect theory, the values of γ and δ range between 0 and 1, with a smaller magnitude indicating a stronger degree of overweighting tail events. Previous studies (Tversky and Kahneman, 1992; Booi et al., 2010) suggest an estimated value of 0.65 for both parameters. However, our findings indicate an even greater degree of probability weighting in both the gain and loss regions. The estimated mean value for γ is 0.228, approximately a third in magnitude as those reported in previous studies. This suggests that the probability weighting in the loss region, as revealed by mutual fund flow data, is stronger than what has been observed in laboratory settings. Furthermore, the mean value of δ for the gain region is even smaller, suggesting that mutual fund investors exhibit a more aggressive overweighting of tail events in the gain region compared to the loss region. The strong probability weighting parameters from our estimation are consistent with the lottery preferences documented in the previous literature (Kumar, 2009 and Akbas and Genc, 2020).

In sum, our revealed preference analysis provides further support for the important influence of prospect theory among mutual fund investors, highlighted by the significant loss aversion and strong probability weighting. In the remaining sections, we revert to the prospect theory values of mutual funds under lab-based parameters given in (8), but our findings remain robust if we adopt the parameter estimates obtained above under the revealed preference analysis.

4.3.4. Prior investment performance and loss aversion

The seminal work of Barberis et al. (2001) highlights the importance of time-varying loss aversion in explaining the equity premium under prospect theory. They argue that investors' degree of loss aversion depends on their prior investment performance. After prior gains, investors become less loss averse: the prior gains will cushion any subsequent loss, making it more bearable. Conversely, after a prior loss, investors become more loss averse: after being burned by the initial loss, investors become more sensitive to additional setbacks.¹⁷ Although significant and well-known, there is limited field evidence for this time-varying pattern of loss aversion.

We provide an empirical test by leveraging the variation in loss aversion through our quarter-by-quarter estimation. To proxy for investors' prior gains, we use the market return in the last quarter as it captures the wealth fluctuations at the market-wide level. The first proxy is the market return downloaded from Kenneth French's website, which represents the value-weighted return of all CRSP firms incorporated in the US and listed on the NYSE, AMEX, or NASDAQ. We use returns from the last quarter to predict loss aversion in the current quarter. Based on Table 8 Panel B Column (1), a one standard deviation increase in the quarterly market return (7.288%) significantly decreases the degree of loss aversion by 0.079, which amounts to 71.82% of the standard deviation of loss aversion (calculated as $7.288/100 \times 1.084/0.11$). The second proxy is the return of the S&P 500 index. Similarly, we observe a significantly negative relation between last quarter's S&P 500 return and the loss aversion parameter inferred from mutual fund flow. Overall, we provide supportive evidence that prior investment performance affects the degree of loss aversion exactly as assumed in the model of Barberis et al. (2001).

5. Prospect theory: A new framework for mutual fund flows

In this section, we investigate whether prospect theory can provide new insights into what drives mutual fund flows. We begin by exploring how investors can intuit TK variations by relying on intuitive and easily accessible indicators that reflect the key components of prospect theory. We then discuss the determinants of fund TK values from the perspectives of both investors and fund management. We find that prospect theory possesses novel and robust predictive power for mutual fund flows beyond existing determinants under both rational and behavioral theories.

5.1. How do investors intuit TK?

When allocating capital to mutual funds, investors may not necessarily compute TK as we do. Instead, they can rely on intuitive indicators reflecting the key principles of prospect theory that correlate with TK. As a result, they behave as if they are choosing funds with high TK values. We hypothesize that one such intuitive indicator is the loss-gain ratio, which reflects the loss-aversion aspect of prospect theory. If investors are influenced by loss aversion, they may assess the attractiveness of a fund by simply weighing total losses against total gains. To test this, we measure a given fund's loss-gain ratio as the absolute value of the sum of its negative excess returns over the previous 60 months (total loss) divided by the sum of positive excess returns over the previous 60 months (total gain). Column (1) of Table 9 shows a significantly negative relation between mutual fund flows and the fund's loss-gain ratio. This supports the idea that real-world investors can intuit variations in TK by tracking simple indicators such as the loss-gain ratio. Moreover, column (2) of Table 9 indicates an asymmetric impact of losses versus gains on fund flows, consistent with the loss aversion feature of prospect theory. Specifically, the coefficient for total losses is twice as large as that for total gains.

¹⁶ We follow the procedure in Roussanov et al. (2021) to exclude institutional funds.

¹⁷ This idea dates back to Thaler and Johnson (1990) who document that when faced with sequential gambles, people are more willing to take risks if they made money on prior gambles than if they lost. This pattern is known as the "house money" effect.

Table 8
Revealed preference analysis: Prospect theory parameters from the field.

Panel A. Estimation of Parameters				
	description	mean	s.e.	99% CI
α	Curvature of the value function	0.745	0.061	[0.576, 0.914]
λ	Loss aversion	1.824	0.110	[1.529, 2.119]
γ	Probability weighting in gain region	0.110	0.028	[0.034, 0.187]
δ	Probability weighting in loss region	0.228	0.041	[0.117, 0.340]

Panel B. Prior Investment Performance and Loss Aversion		
	(1) Loss Aversion(t)	(2) Loss Aversion(t)
Market Return (t-1)	-1.084* (-2.03)	
Return of the S&P 500 Index (t-1)		-1.175** (-2.12)
Adjusted R-Squared	0.018	0.018
N	30	30

This table presents the estimated prospect theory parameters based on the discrete choice model using the mutual fund flows data. We estimate the parameters on a quarterly basis. In Panel A, we report the mean, standard deviation, and 99% confidence intervals constructed from the quarterly estimates of the parameters. In Panel B, we use investment performance in the previous quarter to predict loss aversion in the current quarter. We rely on our quarter-by-quarter estimation to obtain the time series for loss aversion. We use market returns in the last quarter as a proxy for prior investment performance. In column (1), we use the market return downloaded from Kenneth French's website, which represents the value-weighted return of all CRSP firms incorporated in the US and listed on the NYSE, AMEX, or NASDAQ. In column (2), we use the return of the S&P 500 index. The *t*-statistics are based on standard errors corrected by Newey–West with lags of 4 quarters.

Table 9
Gain/Loss ratio and mutual fund flows.

	Dep. Var. Flow	
	(1)	(2)
Loss/Gain ratio	-0.010*** (-2.89)	
Total loss		-0.018*** (-6.19)
Total gain		0.007** (2.31)
Adjusted R-Squared	0.135	0.134
N	522,823	522,823
Fund FE	Yes	Yes
Date FE	Yes	Yes
Controls	Yes	Yes

This table presents the predictive power of a simple loss–gain ratio indicator in forecasting future fund flows. This simple indicator, motivated by the loss-aversion feature of prospect theory, helps us study how investors intuit the variations in TK. First, we calculate Total Loss (Gain) as the sum of negative (positive) excess returns in the previous 60 months. Then we compute the ratio using the absolute value of total loss divided by the total gain. We regress fund flows in the subsequent month on the Loss/Gain ratio (column 1) and Total Loss and Total Gains (column 2). The control variables are identical to those included in column (4) of Table 3. In all specifications, we include both fund and date fixed effects. The sample spans from January 1981 to June 2022. Standard errors are clustered at the fund and date level.

An equally compelling angle to understand how real-world investors intuit variations in TK is probability weighting, another key feature of prospect theory. Investors tend to overweight the probability of extreme and rare events, leading to preferences for lottery-like returns (Barberis and Huang, 2008, Kumar, 2009). In the existing literature, researchers rely on variables such as skewness and maximum return to proxy for lottery-like characteristics. We argue that these variables also serve as easily accessible indicators for prospect theory investors to intuit variations in TK and assess fund attractiveness. This is also the reason that we do not control for fund skewness in our baseline specification, as it is an integral part of prospect theory (Barberis et al., 2016, Barberis et al., 2021).

5.2. Determinants of TK

In previous sections, we present evidence that TK robustly predicts future fund flows. In the following, we discuss the drivers of TK from both the perspectives of mutual fund investors and fund management.

TK quantifies the attractiveness of a fund as perceived by prospect theory investors, based on their mental representation of the fund's return distribution. In our analysis, we follow (Barberis et al., 2016) by using the fund's realized return history as a proxy for this mental representation. Consequently, variations in TK are ultimately driven by fund return distributions (and the associated prospect theory values perceived by investors). In the time-series context, as the return history of a given fund evolves, the mental representation also changes, leading to within-fund variation in TK over time. At a given point in time, different funds will have distinct return distributions, resulting in different prospects of future gains and losses, thereby generating cross-sectional variation in TK.

Since TK is derived from historical returns, it is not surprising that it closely correlates with the statistical moments of fund returns, as illustrated in Table 10. Funds with higher TK values tend to have higher average past returns, lower volatility, and higher skewness. Moreover, it is important to emphasize that prospect theory, as a psychologically realistic framework, captures many nuanced aspects of investor decision-making that extend beyond simple statistical moments (see Barberis, 2013). For example, loss aversion, a key aspect of prospect theory, is not adequately reflected by statistical moments. In Table 10, we confirm that these critical features of prospect theory, such as loss aversion, influence TK variations, as indicated by intuitive measures like the loss–gain ratio.

It is also important to understand how decisions by mutual fund managers affect the variations in TK. To gain further insight into what drives TK's variation across funds, we examine several holdings-based measures to understand the relationship between a fund's portfolio characteristics and its TK value. To obtain the holdings data, we merge the CRSP Mutual Fund Database with Thomson Reuter's Mutual Fund Holdings Database using link tables from MFLINKS.

To begin with, we investigate whether funds have higher TK values because they tend to hold more high-TK stocks. Specifically, we construct a holdings-based TK measure for each fund by first calculating

Table 10
Determinants of TK — Investor's Perspective.

	Dep. Var. TK	
	(1)	(2)
Loss/Gain ratio	−0.012*** (−7.17)	
Total loss		−0.020*** (−11.19)
Total gain		0.006*** (3.87)
Cumulative returns (60 m)	0.008*** (15.45)	0.005*** (10.03)
Return volatility	−0.497*** (−32.30)	−0.244*** (−3.75)
Skewness	0.005*** (14.28)	0.006*** (16.62)
Adjusted R-Squared	0.819	0.825
N	522,823	522,823
Fund FE	Yes	Yes
Date FE	Yes	Yes
Controls	Yes	Yes

This table examines the determinants of TK from the perspective of mutual fund investors. The dependent variable is the fund's TK value. The explanatory variables include losses and gains from the fund's past return distribution, as well as three statistical moments of the return distribution: cumulative returns, return volatility, and skewness. Total Loss (Gain) is defined as the sum of negative (positive) excess returns over the previous 60 months. The Loss/Gain Ratio is the ratio of the absolute value of the total loss to the total gain. All these variables are estimated using fund returns from the preceding 60 months. The additional control variables are identical to those included in column (4) of Table 3. In all specifications, we include both fund and date fixed effects. The sample spans the period from January 1981 to June 2022. Standard errors are clustered at the fund and date level.

the stock-level TK using the stock's historical monthly returns. We then calculate the average of these TK values across the stocks within a fund's holdings, using portfolio weights. We perform the Fama-MacBeth regression of the fund-level TK (our main measure in the paper) on the holding-based TK measure, while controlling for a set of standard fund characteristics. As shown in column (1) of Table 11, the coefficient of the holdings-based TK measure is significant and positive, indicating that the fund's TK is partly driven by its selection of stocks with high TK values. Moreover, we construct an alternative holdings-based TK measure using the current portfolio, deriving the past 60-month returns based on current portfolio weights and the historical returns of the constituent stocks. Using this alternative measure, we again confirm a significant relationship between fund-level TK and holdings-based TK, as demonstrated in column (2) of Table 11.

We then explore whether funds with high TK values tend to hold more diversified portfolios. *Ex ante*, the impact of diversification on a fund's TK is unclear. On the one hand, a more diversified portfolio tends to have lower return volatility, which leads to a higher TK value. On the other hand, diversification reduces the portfolio's skewness, which could lower TK. We measure the degree of diversification using the Herfindahl index of portfolio weights of the fund holdings (hereafter referred to as Holding HHI). The coefficient for Holding HHI in column (3) of Table 11 is negative and significant, indicating that less diversified funds (those with high HHI), which tend to have higher volatility, are associated with lower TK.

Furthermore, we examine whether funds with higher TK values are more likely to adjust their risk exposures. Following Huang et al. (2011), we construct a holdings-based measure of risk shifting by gauging the difference between a fund's current holdings volatility and its past realized volatility. Column (4) of Table 11 shows that funds with more risk shifting tend to have higher TK values. Considering the positive flow-TK relationship, this finding is consistent with existing literature that indicates mutual funds have incentives to strategically adjust their risk levels to attract additional fund flows (Huang et al., 2011).

Table 11
Determinants of TK — Fund management perspective.

	Dep. Var. TK			
	(1)	(2)	(3)	(4)
Average stock TK	0.164*** (4.74)			
TK of held portfolio		0.405*** (17.69)		
Holding HHI			−0.009*** (−2.87)	
Risk shifting				0.121*** (5.36)
Average Adj. Rsq.	0.363	0.546	0.313	0.299
N	453,562	377,486	400,662	444,263
Controls	Yes	Yes	Yes	Yes

This table presents the determinants of TK from the fund management perspective using holdings-based measures. The dependent variable is the fund's TK value. Average Stock TK is calculated as the weighted average of the stock TK values using the portfolio weights of the fund. The TK of the Held Portfolio is determined by calculating the TK using the returns of the portfolio based on the fund's current holdings. Holding HHI is the Herfindahl-Hirschman Index calculated based on the portfolio weights of the funds. Risk Shifting is defined as the difference between the volatility calculated from the past returns of the currently held portfolio and the fund's realized return (Huang et al., 2011). The control variables are identical to those included in Column (4) of Table 3. The sample covers the period from January 1981 to June 2022. All regressions are estimated using the Fama-MacBeth method. Standard errors are clustered at the fund and date levels.

5.3. Controlling for other fund flow drivers

We differentiate prospect theory from known determinants of mutual fund flows, such as Morningstar ratings and behavioral drivers including return extrapolation, salience, and the maximum or skewness of returns. In the presence of these controls, prospect theory demonstrates both novel and robust predictive power for mutual fund flows.

5.3.1. Morningstar ratings and expected utility

One of the most prominent drivers of mutual fund flows is Morningstar ratings. Extant literature documents strong evidence that investors tend to chase mutual funds based on Morningstar ratings (see Reuter and Zitzewitz, 2021, Ben-David et al., 2022, and Del Guercio and Tkac, 2008). Morningstar rates funds based on their past cumulative returns, adjusted in a way that penalizes high volatility, according to the expected utility framework.

Although TK is also constructed as a function of past fund returns, there are important differences that distinguish it from Morningstar ratings. Unlike TK, Morningstar ratings do not incorporate prospect theory features such as loss aversion and overweighting of tail events. Still, we carefully control for Morningstar ratings. First, since the unit of observation in our baseline regression is at the fund level, we construct the fund level 5-year rating by calculating a weighted average of the 5-year ratings from share classes with available rating data, where the weight of each share class is its TNA divided by the aggregate TNA of all share classes. As shown in column (1) of Panel A in Table 12, the relation between TK and fund flow remains robust and significant after controlling for the Morningstar ratings. Moreover, we also estimate the flow-TK regression at the share class level, where TK and control variables (including Morningstar ratings) are measured at the share class level. The results are very similar, as shown in column (3) of Panel A in Table 12.

To further rule out the concern that TK's predictive power for fund flows is driven by Morningstar ratings, we follow the approach in Barber et al. (2016) and replace the date fixed effect with the date-rating interaction fixed effect.¹⁸ In a given period, we condition on funds with the same Morningstar ratings and examine how variations

Table 12
Controlling for morningstar ratings and morningstar risk adjusted returns.

Panel A. Control for Ratings				
Sample:	Dep. Var. Flow			
	Fund-level (1)	(2)	Share-class level (3)	(4)
TK	0.140*** (3.97)	0.145*** (4.59)	0.119*** (3.77)	0.120*** (4.51)
5Y MS rating (weighted)	0.006*** (32.69)		0.006*** (41.23)	
Adjusted R-Squared	0.162	0.170	0.151	0.156
N	573,599	573,599	1,240,149	1,240,149
Fund FE	Yes	Yes	Yes	Yes
Date FE	Yes	No	Yes	No
Rating × Date FE	No	Yes	No	Yes
Controls	Yes	Yes	Yes	Yes

Panel B. Control for MRAR				
	Dep. Var. Flow			
	Fund-level (1)		Share-class level (2)	
TK	0.349*** (9.50)		0.387*** (11.14)	
MRAR	0.043*** (5.43)		0.001*** (6.16)	
Adjusted R-Squared	0.132		0.130	
N	623,496		1,268,242	
Fund FE	Yes		Yes	
Date FE	Yes		Yes	
Controls	Yes		Yes	

This table presents the predictive power of prospect theory value in forecasting future fund flows while controlling for Morningstar Ratings (Panel A) and Morningstar Risk Adjusted Return, or MRAR (Panel B). In Panel A, we control for the 5-year Morningstar rating and estimate the regressions at the fund level (column (1)) and share-class level (column (3)). The fund-level 5-year rating is aggregated from the share-class level 5-year ratings, weighted by the fraction of fund assets. In columns (2) and (4), we include the Rating × Date fixed effects following (Barber et al., 2016). In Panel B, we control for the Morningstar Risk Adjusted Return (MRAR), which is calculated using the past five years' monthly returns based on a power utility function. In all models, the control variables are identical to those included in column (4) of Table 3. The sample spans from January 1981 to June 2022. Standard errors are clustered at fund and date levels.

in TK predict flows in the next period. This analysis helps us tease out the marginal effect of TK while controlling for the effect of Morningstar ratings. As shown in columns (2) and (4) of Panel A in Table 12, TK remains positive and statistically significant at both the fund and share class levels.

In addition to Morningstar ratings, we control for the Morningstar Risk Adjusted Returns (MRAR), which is calculated using a variant of the power utility as follows:

$$MRAR_i^T = \left[\frac{1}{T} \sum_{t=1}^T (1 + r_{i,t} - r_t^f)^{-\gamma} \right]^{-\frac{12}{\gamma}} - 1,$$

with the risk aversion coefficient $\gamma = 2$. As shown in Panel B of Table 12, both TK and MRAR are positive and significant in predicting future fund flows. Controlling for MRAR also provides a comparison between the *Expected Utility* framework and the prospect theory framework.¹⁹ As mutual fund flows reflect the collective choices of investors, this suggests that some mutual fund investors have expected utility while others are prospect theory decision-makers.

5.3.2. Return extrapolation and salience theory

Greenwood and Shleifer (2014) and Barberis et al. (2015) utilize survey data on beliefs to demonstrate that investors exhibit a decaying memory, with more recent returns exerting a greater influence on their return expectations compared to distant ones. In line with this, we adopt the memory-decaying structure estimated by Barberis et al. (2015) and quantify return extrapolation as a control variable using a weighted sum of past monthly fund returns, where the weights decline exponentially.²⁰

Bordalo et al. (2012) argue that due to cognitive limitations, decision-makers' attention is drawn to the most salient attributes of the options they face. Consequently, these salient attributes are overweighted in their decisions, while nonsalient attributes are often neglected. Cosemans and Frehen (2021) provide empirical evidence for the salience-theory-based asset pricing model of Bordalo et al. (2013). We adopt the same methodology to measure the salience theory value of a fund. Specifically, we calculate a weighted sum of its monthly returns over the past 12 months, with the weight reflecting the degree of salience as defined in Bordalo et al. (2013). This ensures that our measure aligns precisely with the concept of salience in the literature.²¹

¹⁸ Following Barber et al. (2016), we convert weighted 5-year rating at fund-level into 5 categories: [0, 1.5), [1.5, 2.5), [2.5, 3.5), [3.5, 4.5), [4.5, 5]. For the share-class level, we directly use the 5-year rating.

¹⁹ In Internet Appendix Table B9, we control for the expected utility value under standard power utility and obtain similar results.

²⁰ Greenwood and Shleifer (2014) also estimate a memory-decaying structure using quarterly returns. We find that the results remain robust under both specifications.

²¹ For detailed procedures to calculate the return extrapolation value and the salience theory value, please refer to Internet Appendix B.

Table 13

Prospect theory and fund flows: Controlling for behavioral drivers of fund flows.

	Dep. Var. Flow			
	(1)	(2)	(3)	(4)
TK	0.247*** (6.70)	0.354*** (10.59)	0.327*** (9.80)	0.295*** (6.58)
EX	1.177*** (15.90)			
ST		0.106*** (12.25)		
MAX			9.566*** (13.25)	
Skewness				0.002*** (3.78)
Adjusted R-Squared	0.152	0.140	0.141	0.134
N	859,562	859,562	859,562	522,823
Fund FE	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes

This table compares the predictive power of prospect theory and other behavioral drivers of fund flows. The values of prospect theory (TK) are estimated using fund returns from the preceding 60 months. The values of return extrapolation (EX) are estimated using fund returns from the preceding 60 months, adopting the memory-decaying structure documented in [Greenwood and Shleifer \(2014\)](#). The values of salience theory measurement (ST) are based on fund returns in the past 12 months, following [Bordalo et al., 2013](#). The values of max returns (MAX) are based on the maximum value of the market-adjusted fund returns in the past 12 months, as outlined in [Akbas and Genc \(2020\)](#). The values of skewness (Skewness) are based on the distribution of fund returns in the past 60 months. In all specifications, we include both fund and date fixed effects. The control variables are identical to those included in column (4) of [Table 3](#). The sample spans the period from January 1981 to June 2022. Standard errors are clustered at the fund and date level.

Table 14

Prospect theory and fund flows: Institutional versus Retail funds.

	Dep. Var. Flow		
	(1)	(2)	(3)
TK	0.383*** (11.27)	0.390*** (11.36)	0.368*** (10.50)
Institutional × TK		−0.033** (−2.22)	
Institutional		−0.000 (−0.32)	
Retail × TK			0.024* (1.95)
Retail			−0.001 (−1.51)
Adjusted R-Squared	0.138	0.138	0.138
N	859,562	859,562	859,562
Fund FE	Yes	Yes	Yes
Date FE	Yes	Yes	Yes
Controls	Yes	Yes	Yes

This table presents the heterogeneous predictive power of prospect theory in forecasting flows of funds dominated by different types of investors. The values of prospect theory (TK) are estimated using fund returns from the preceding 60 months. We use two variables as proxies for investor types: Institutional, the dummy variable for institutional funds, and Retail, the dummy variable for retail funds. In column (1), we report the baseline regression results. In column (2), we report the result from the regression that includes the dummy variable for institutional funds. In column (3), we report the result from the regression that includes the dummy variable for retail funds. In all specifications, we include both fund and date fixed effects. The control variables are identical to those included in column (4) of [Table 3](#). The sample spans the period from January 1981 to June 2022. Standard errors are clustered at the fund and date level.

Table 15

Prospect theory and fund flows: Effect of investor sophistication.

	Dep. Var. Flow		
	(1)	(2)	(3)
TK	0.383*** (11.27)	0.363*** (10.51)	0.392*** (11.51)
Broker Sold × TK		0.038*** (3.22)	
Broker Sold		0.001 (1.10)	
Direct Sold × TK			−0.036*** (−2.76)
Direct Sold			−0.004*** (−5.41)
Adjusted R-Squared	0.138	0.138	0.138
N	859,562	859,562	859,562
Fund FE	Yes	Yes	Yes
Date FE	Yes	Yes	Yes
Controls	Yes	Yes	Yes

This table presents the heterogeneous predictive power of prospect theory in forecasting fund flows dominated by investors with varying levels of sophistication. The values of prospect theory (TK) are estimated using fund returns from the preceding 60 months. Following [Barber et al., 2016](#), we utilize two variables to reflect investor sophistication: Broker Sold, a dummy variable for funds distributed mainly through brokers, and Direct Sold, a dummy variable for funds sold directly. In column (1), we report the baseline regression results. In column (2), we present the results of regressions that include the dummy variable for funds distributed through brokers. In column (3), we provide the results of regressions that include the dummy variable for directly sold funds. In all specifications, we include both fund and date fixed effects. The control variables are identical to those included in column (4) of [Table 3](#). The sample spans the period from January 1981 to June 2022. Standard errors are clustered at the fund and date level.

Table 16

Prospect theory and fund flows: Time-varying predictive power.

	Dep. Var. Flow	
	(1) Investor sentiment	(2) NBER recessions
TK	0.082** (2.55)	0.084*** (2.70)
TK × High sentiment	0.074* (1.71)	
High sentiment	0.002* (1.67)	
TK × Recession		−0.099** (−2.15)
Recession		−0.007*** (−3.62)
Adjusted R-Squared	0.113	0.113
N	766,617	766,617
Fund FE	Yes	Yes
Date FE	No	No
Controls	Yes	Yes

This table examines the time-varying predictive power of prospect theory for future fund flows. The values of prospect theory (TK) are estimated using fund returns from the preceding 60 months. We consider two indicators for episodes: the investor sentiment indicator proposed in [Baker and Wurgler \(2006\)](#) and the recession indicator (Recession) defined by the NBER. We define a dummy variable for high investor sentiment (High Sentiment) when the sentiment level exceeds the 75th percentile of the investor sentiment index within the in-sample data. In column (1), we report the results of regressions that include the dummy variable for episodes of high investor sentiment. In column (2), we report the results of regressions that include the dummy variable for recession episodes. In all specifications, we include both fund and date fixed effects. The control variables are identical to those included in column (4) of [Table 3](#). The sample spans the period from January 1981 to June 2022. Standard errors are clustered at the fund and date level.

We add the return extrapolation and salience measures as controls to the baseline flow regression (10) and report the results in the first two columns of Table 13. The predictive power of prospect theory remains strong and significant. Return extrapolation and salience measures also have significant predictive power for mutual fund flows, even after controlling for CAPM alpha as well as prospect theory value. This suggests that extrapolative belief and salience theory play important roles in mutual fund investors' decision-making, but they cannot explain the significantly positive relation between mutual fund flows and prospect theory value.

5.3.3. Max returns and skewness

One important implication of the prospect theory is the lottery preference (Barberis and Huang, 2008). Investors with such preferences may chase mutual funds that exhibit lottery-like features, such as past return skewness and max returns. For example, Akbas and Genc (2020) show that maximum returns significantly predict future fund flows because investors prefer funds that have extremely positive payoffs or positive skewness. In essence, maximum return and skewness preference are consistent with the probability weighting feature of prospect theory. As we have shown in Section 4.1.5, however, all features of prospect theory contribute to its explanatory power for mutual fund flows. Hence, we expect TK to remain a significant determinant of fund flows after controlling for maximum returns or skewness.

Following Akbas and Genc (2020), we include the maximum style-adjusted returns from the preceding 12 months as an additional control in our baseline flow regression (10). Table 13 column (3) confirms the significant predictive power of max return for fund flows. Consistent with our conjecture, TK is still significantly and positively related to fund flows in the presence of maximum returns. The coefficient of TK only drops slightly in magnitude. Our results also remain robust after controlling for the fund skewness estimated using the preceding 60 monthly returns, as shown in column (4) of Table 13.

Overall, the findings in this section confirm the novelty and robustness of the explanatory power of prospect theory for mutual fund investors' decision-making.

6. Bounded-rational drivers of fund flows

As a psychologically grounded framework, prospect theory is widely recognized for capturing non-fully rational behaviors. In this section, we present further evidence supporting the presence of less-than-fully-rational investor behavior in picking mutual funds within the framework of prospect theory.

6.1. Investor heterogeneity

Our previous analysis considers all types of funds together. In this subsection, we investigate the heterogeneous effects of prospect theory across investor clienteles with different levels of sophistication. We hypothesize that the effect of TK on mutual fund choices is stronger for less sophisticated investors (e.g., retail investors) because they are more influenced by prospect theory.

6.1.1. Institutional and retail funds

Mutual funds are sold in several share classes. Some share classes are sold to institutional investors, while others target retail investors. Empirically, we classify a fund as an institutional (retail) fund if more than 75% of its TNA falls into institutional (retail) share classes. Since institutional investors are less prone to behavioral biases and less likely to be influenced by prospect theory, we expect prospect theory to provide a stronger explanatory power for retail fund flows. We test this hypothesis by interacting institutional or retail fund indicators with TK. We present the results in Table 14. For column (1), we repeat the baseline regression associated with Table 3 for comparison. For column (2), we interact TK with the indicator for institutional funds.

The coefficient of the interaction term is negative and significant. Similarly, for column (3), we interact TK with the indicator for retail funds and find the interaction term to be positive and significant. These two results are consistent with each other. They suggest a stronger (weaker) effect of funds' prospect theory value on fund flows for funds primarily targeting retail (resp. institutional) investors. This confirms our conjecture that the explanatory power of prospect theory for fund flows depends on the sophistication of the investor clientele.

6.1.2. Distribution channel

Mutual funds differ with respect to their distribution channels. In the US, mutual funds are either sold directly to investors or distributed via intermediaries such as brokers and financial advisors. Previous evidence reveals distinct features of investors across distribution channels. Chalmers and Reuter (2012) find that investors who buy mutual funds through brokers are "younger, less highly educated, and less highly paid". Christoffersen et al. (2013) document that flows to broker-sold funds are influenced by payments from fund companies to brokers. Moreover, Bergstresser et al. (2008) and Guercio and Reuter (2014) both show that broker-sold mutual funds do not seem to provide additional benefits relative to directly sold mutual funds. Based on the extant literature, investors of broker-sold funds appear less sophisticated than investors of directly sold funds. Thus, we expect the effect of TK on fund flows to be stronger for broker-sold funds. We classify broker-sold funds following the procedure described in Sun (2014).²²

The results are reported in Table 15. Column (1) repeats the baseline results. We interact TK with an indicator for broker-sold funds in column (2) and with an indicator for direct-sold funds in column (3). The interaction term between the broker-sold dummy and TK is positive and significant, while the interaction term between the direct-sold dummy and TK is negative and significant. The results suggest that prospect theory is more powerful at explaining flows of broker-sold mutual funds where investors are less sophisticated. This further supports our hypothesis in this subsection.

6.2. Investor sentiment

The underlying foundations of prospect theory are closely related to judgment heuristics, such as narrow framing, as posited in the psychology literature. To the extent that prospect theory captures non-fully rational behavior, we expect the effect of prospect theory on fund flows to be stronger when investors are more susceptible to the influence of "animal" spirits. To test this hypothesis, we use the investor sentiment index proposed in Baker and Wurgler (2006). We define episodes of high investor sentiment when the sentiment level exceeds the 75th percentile of the investor sentiment index over our sample period. We include an interaction term between TK value and the indicator for high investor sentiment. Consistent with our expectation, Table 16 column (1) shows a positive and significant coefficient for the interaction term, indicating that the predictive power of prospect theory for fund flows intensifies when market sentiment is high.²³ Similarly, in column (2), we include an interaction term between TK value and the NBER recession dummy. We find that the predictive power of TK drops significantly during recessions.

²² A broker-sold fund holds at least 75% of its TNA in share classes that charge front-end loads, back-end loads, or 12b-1 fees greater than 25 bps. A directly sold fund holds at least 75% of its TNA in share classes that do not charge front-end loads, back-end loads, or 12b-1 fees.

²³ In untabulated results, we find that the positive relationships between fund flow and each individual feature of prospect theory (i.e., LA, CC, PW) documented in Table 5 all become stronger (more positive) during periods of high investor sentiment.

Table 17
TK values and subsequent fund performance.

Panel A. TK values and fund performance			
	(1) 1 month	(2) 3 months	(3) 12 months
TK	−0.014 (−0.39)	−0.053 (−0.52)	−0.380 (−1.13)
Adj. Rsq.	0.319	0.319	0.359
N	859,592	845,089	782,471
Controls	Yes	Yes	Yes

Panel B. TK-Driven flows and fund performance			
	(1) 1 month	(2) 3 months	(3) 12 months
$\widehat{flow}$	−0.001** (−2.07)	−0.003* (−1.87)	−0.008 (−1.56)
Adj. Rsq.	0.309	0.310	0.349
N	843,151	828,864	767,360
Controls	Yes	Yes	Yes

Panel C. Non-TK-Driven flows and fund performance			
	(1) 1 month	(2) 3 months	(3) 12 months
u	0.006*** (5.62)	0.009*** (4.27)	0.010** (2.31)
Adj. Rsq.	0.310	0.310	0.349
N	843,151	828,864	767,360
Controls	Yes	Yes	Yes

This table presents the results of Fama–MacBeth regressions. The dependent variables are the future fund alphas based on the Fama–French–Carhart four-factor model, with horizons ranging from 1 month to 12 months. The key independent variables are the fund's TK value in Panel A, TK-driven flows in Panel B, and non-TK-driven flows in Panel C. The values of prospect theory (TK) are estimated using fund returns from the preceding 60 months. TK-driven flows and non-TK-driven flows are estimated, respectively, from the fund-level regressions $Flow_{i,t} = a + bTK_{i,t-1} + u_{i,t}$ over the expanding window. The control variables are identical to those included in column (4) of Table 3. The sample spans the period from January 1981 to June 2022. We report standard errors corrected by Newey–West with lags of 12 months.

6.3. Flows and subsequent fund performance

Our main finding so far is that funds with high TK values attract high fund flows, especially for less sophisticated investor clientele and during periods of high investor sentiment. If higher TK value is associated with superior subsequent fund performance, then investors are smart to chase high TK value funds. If the opposite holds, then the TK-chasing behavior would be wealth-destroying and dumb.

In Panel A of Table 17, we run Fama–MacBeth regressions of funds' future four-factor alphas on TK value. We compute a fund's future alpha as follows. First, for each fund, we estimate its time-varying four-factor betas by regressing the fund's excess returns on the four factors over an expanding window. Second, we use the estimated betas to compute the future four-factor alpha. For example, to calculate the four-factor alpha for period $t + 1$, we use the estimated betas from the end of period t , along with the realized fund excess returns and the values of the four factors during period $t + 1$. We check the predictive relationship between TK and alphas over multiple horizons, extending up to 12 months. The results do not lend any support for a significant and positive relationship between fund TK value and its future alphas. In fact, there is some evidence that funds with high TK values tend to underperform subsequently.

Fund alpha depends crucially on fund size in addition to managerial skill (Berk and Van Binsbergen, 2015). High TK funds could attract excessive fund flows that end up hurting the fund performance due to decreasing returns to scale in deploying managerial skills (Berk and Green, 2004). Thus, we hypothesize that the effect of TK on future

fund performance could arise from the price impact of capital flows from prospect theory investors. To verify this conjecture, we decompose fund flows into a TK-driven component and a non-TK-driven component by projecting fund flows on lagged TK. We then re-run the Fama–MacBeth regression in Panel A of Table 17 with future alphas as the dependent variable and the decomposed flow components as independent variables. As reported in Panel B and Panel C of Table 17, TK-driven flows predict future alphas negatively, while non-TK-driven flows predict future alphas positively. Both results are statistically significant. For instance, as shown in column (2) of Table 17, a one-standard-deviation increase (0.255) in the TK-driven flow results in a reduction of future alpha (annualized) by 31 basis points (calculated as $0.255 \times (-0.003) \times 4$). This finding suggests that the effect of TK on future fund performance operates through flows, and TK-chasing behavior is detrimental to performance. Interestingly, the remaining mutual fund flows – those not driven by prospect theory – appear to represent “smart money”, as they are positively associated with future fund alphas.

7. Conclusion

Investors select among risky assets based on their assessments of the return distributions through the lens of their preferences or utility functions. These choices provide researchers with valuable insights into the underlying preferences. Building on this rationale, we utilize mutual fund flows to empirically test theories that seek to explain investor behavior. Our focus is on prospect theory. Widely recognized as a psychologically realistic model of preferences in laboratory environments, prospect theory has garnered growing interest from finance researchers. We provide field evidence on the practical relevance of prospect theory for mutual fund investors. Moreover, we conduct a revealed preference analysis to obtain estimates of prospect theory parameters from market participants' capital allocations to mutual funds.

Our empirical findings provide compelling evidence supporting the role of prospect theory in driving investors' choices among mutual funds: funds whose past returns generate higher prospect theory values attract significantly larger future flows. In particular, for an average fund in our sample, with a Total Net Assets of \$1,533 million, a one-standard-deviation increase in TK results in a flow increment of \$11.16 million. Each feature of prospect theory plays an indispensable role in shaping this predictive pattern. Our account-level evidence corroborates these results. Furthermore, our field-based prospect theory parameter estimates closely match those from experimental studies. This further validates prospect theory's broad relevance in the mutual fund setting.

Our results demonstrate that prospect theory provides a novel framework for understanding mutual fund flows. The predictive power of prospect theory remains robust even after controlling for fund performance measures and other known determinants of flows such as Morningstar ratings. Our findings align well with boundedly rational mutual fund investors. Specifically, the effect of prospect theory value on fund flows is more pronounced among retail funds and broker-sold mutual funds where investors are less sophisticated. It is also stronger during periods of high investor sentiment. Additionally, we find that prospect theory-driven flows are followed by subsequent fund underperformance. This points to a “dumb money” effect associated with investors exhibiting prospect theory preference. Overall, this paper highlights the importance of prospect theory for capital allocation in the mutual funds market. Similar studies can be conducted to better understand the demand system in other financial markets.

CRedit authorship contribution statement

Bing Han: Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Conceptualization. **Pengfei Sui:** Writing – original draft, Investigation, Formal analysis, Data curation, Conceptualization. **Wenhao Yang:** Writing – original draft, Validation, Investigation, Formal analysis, Conceptualization.

Table A.18
Definition of Variables.

Variables	Description
Flow	Defined as the percentage growth of new assets, following the fund-flow literature
New Subscriptions	Dollar amount of total new subscriptions in each month scaled by a fund's total net assets at the end of the preceding month
Redemptions	Dollar amount of redemption in each month scaled by a fund's total net assets at the end of the preceding month
TK	Defined as a fund's prospect-theory value based on fund returns in the preceding 60 months. "TK" value represents fund value as specified in Tversky and Kahneman (1992), the paper that first introduced cumulative prospect theory.
LA	Defined as a fund's prospect-theory value, featuring loss aversion while turning off probability weighting and the concavity/convexity feature in the value function.
CC	Defined as a fund's prospect-theory value, featuring concavity/convexity while turning off loss aversion and probability weighting in the value function.
PW	Defined as a fund's prospect-theory value, featuring probability weighting while turning off loss aversion and the concavity/convexity feature in the value function.
CAPM Alpha	The intercept from the time-series regression of monthly fund returns on market excess returns using the preceding-60-months window
FF4 Alpha	The intercept from the time-series regression of monthly fund returns on Fama–French–Carhart 4-factors using the preceding-60-months window
Market Loading, SMB Loading, HML Loading, MOM Loading	The coefficients of Market, SMB, HML, and Momentum factors from the time-series regression of monthly fund returns on Fama–French–Carhart 4-factors using the preceding-60-months window
FF4 R Squared	The R-squared from the time-series regression of monthly fund returns on Fama–French–Carhart 4-factors using the preceding-60-months window
Ln(Age)	Logarithmic form of fund age in years
Ln(TNA)	Logarithmic form of a fund's total net assets
Expense Ratio	A fund's expense ratio
Turnover Ratio	A fund's turnover ratio
Return Volatility	Standard deviation of monthly fund returns in the preceding 60 months
Cumulative Returns (60 m)	Cumulative monthly fund returns in the preceding 60 months

This table presents the definitions of the main variables in our study.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Variable definitions

See Table A.18.

Appendix B. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jfineco.2025.104221>.

Data availability

Prospect Theory in the Field (Reference data) (Mendeley Data)

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